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NARRATIVE AND ARGUMENTATION ACROSS DISCOURSE TYPES: A COMPARATIVE STUDY OF HUMAN AND LLM ANNOTATION

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ABSTRACT

This thesis investigates how narrativity and discourse type influence the identification of explicit and implicit argumentative components by human annotators and Large Language Models. The study compares two discourse types: STORAL folktales, where argumentative meaning is usually implicit and embedded in narrative events, and TED Talk transcripts, where argumentation is generally more explicit but often supported by narrative elements such as anecdotes, examples, and personal experiences.

The research combines human annotation, prompt-based LLM annotation, quantitative evaluation, and qualitative error analysis. A final corpus of 40 texts was used, consisting of 20 STORAL stories and 20 TED Talk transcripts. The annotation framework focused on major claims, claims, and premises, while also distinguishing between explicit and implicit argumentative components. For TED talks, an additional narrative layer was used to classify narrative spans according to their rhetorical function. Three instruction-tuned LLMs, Gemma, Mistral, and Qwen, were evaluated against the human annotations.

The results show that narrativity increases the interpretive complexity of argument identification. In the TED corpus, human-human agreement was high for explicit argumentation, with an average soft Gamma score of 0.900, but much lower for the narrative-function layer, with an average score of 0.105. This suggests that explicit argumentative components were more consistently identifiable than the rhetorical functions of narrative spans. The LLMs did not reach human-level agreement in the TED argumentation layer and often missed annotations, selected overly broad spans, or confused major claims, claims, and premises. In the STORAL corpus, the models often produced plausible moral interpretations, especially for implicit major claims, but struggled to reconstruct the supporting premises that connected story events to those morals.

Overall, the research shows that narrative affects argument extraction not only by making argumentative meaning more implicit, but also by increasing the number of possible interpretations of the same textual material. The findings suggest that LLMs can support the analysis of argumentative and narrative discourse, but that their outputs require careful evaluation, especially when argumentative meaning is implicit, narratively embedded, or rhetorically ambiguous.

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PREFACE

If you were to ask my friends, they could tell you word for word how proud I am of myself for deciding to dip my toes into the world of coding, coming from someone who always thought she wasn't smart enough for it, but always wished she knew more about it. Thinking back on the first year as a Bachelor student, Master version of Luiza has grown into a more organized, disciplined and level-headed version of herself, who always believed she can do whatever she puts her mind to. With that being said, writing the thesis was as much a pleasure as it was a challenge and these past few months really put to test these new found characteristics.

Many stressful library days and many quiet long nights were spent in front of the screen, trying to understand why my code was not running or giving me errors. Many energy drinks and coffees were consumed to keep me awake (although I think I am immune to caffeine at this point), and many mini heart attacks were had because my laptop (that is hanging on by a thread) sounded like it was going to take off and would freeze at minute 50 out of 55 when running my results.

None of this could have been done if it weren't for the support that I was so lucky to have from my people. First I would like to thank my parents, who have been supporting me through the screen, and who have been so patient when I was excitedly explaining my topic and every step of the process, despite them getting lost in all the details. Same goes for my little sister who I am so lucky to live again with after being apart for 3 years, and who made sure I ate when I forgot to because I was too busy working. The roles reversed and now she took care of me, just like I did when we were little kids. I would also like to thank my boyfriend, who despite also having to deal with writing his own thesis, was there for support when I needed words of encouragement. And of course, I cannot forget my friends, who spent days with me in the library or cafes working together, and always managed to make me laugh when I was too tired or stressed to think. I am the upmost grateful for having such a wonderful support system.

I obviously have to thank my supervisors, for giving me the constructive feedback that helped improve the final version of this thesis, and for putting up with my last minute requests for meetings. Lastly, it cannot go unnoticed, the tremendous amount of help I got from the PhD student that was assigned to my case. She has been a huge support system throughout the whole process, by answering my not so smart questions anytime, without judgement, for spending countless hours annotating with me when she was busy with her own projects, and for always checking in on me.

Thank you

1 | INTRODUCTION

Argumentative meaning is not always expressed through direct claims and clearly stated reasons. While some texts present arguments explicitly through structured reasoning, others communicate argumentative meaning indirectly through narrative elements such as events, characters, actions, consequences, and implied lessons. This distinction is especially important for texts in which narrative plays a central role, because the main claim or moral of a text may not appear as a direct statement, but may instead need to be inferred from the development of the story.

This thesis investigates the relationship between narrativity and argumentative structure across different discourse types. More specifically, it examines how explicit and implicit argumentative components, such as major claims, claims, and premises, are identified in fictional moral narratives and in real-world public talks. The main focus is therefore not persuasion as an effect on readers, but the way argumentative meaning is expressed, structured, and interpreted in texts with different degrees and functions of narrativity.

Narrativity is central to this study because it should not be understood as a simple binary distinction between narrative and non-narrative texts. [Pianzola \(2018\)](#) argues that narrative can be approached as a complex system, where narrativity emerges through the interaction between discourse, audience, and context. This perspective is useful for the current study because it allows narrativity to be understood as a matter of degree and function rather than as something that is either fully present or fully absent. Therefore, public talks, or TED talks, should not be treated as having no narrative role. Instead, they can be understood as argumentative discourse in which narrative elements appear locally and serve rhetorical or argumentative functions. Folktales, by contrast, are highly narrative texts in which narrative structure can be seen through the whole text.

The relationship between narrative and argumentation has often been treated as a distinction between storytelling and reasoning. However, previous work suggests that these two worlds frequently overlap. [Üzelgün et al. \(2017\)](#) argue that narrative and argumentative perspectives are often combined in everyday discourse, while [Tindale \(2017\)](#) shows that narratives can contribute to argumentation when argument is understood in a broader and more dynamic way. From this perspective, stories can function as examples, evidence, background, or implied conclusions. This is particularly relevant for fictional moral stories or folktales, where the argumentative meaning is often not stated directly, but embedded in the sequence of events and the consequences of a character's actions.

The comparison between fictional moral stories and public talks is useful because these discourse types express narrative and argumentation differently. Stories are highly narrative texts that typically communicate moral or social meaning through story events and character actions. In these texts, the reader is often expected to infer a broader moral or conclusion from the plot, a process related to the connection between story plots and implied morals discussed by [Guan et al. \(2022\)](#). TED talks, on the other hand, usually present arguments more explicitly through claims, reasoning, and evidence. However, they also frequently include anecdotes, personal experiences, examples, and background stories that support or frame the argument. The contrast in this thesis is therefore not between narrative and non-narrative discourse, but between two discourse types in which narrative and argumentation interact differently.

This distinction is important for computational discourse analysis and argument mining. Existing argument mining research mostly focused on texts where argumentative components, such as claims and premises, are relatively explicit and can be annotated as textual spans (Stab and Gurevych, 2014a, 2017). However, this becomes more difficult when argumentative meaning is implicit or narratively embedded. In a moral story, for example, the major claim may correspond to the lesson of the story, but that lesson may not be directly stated in the text. Instead, readers must identify the story events that support the interpretation and formulate the implicit argumentative meaning themselves.

More recent computational work has begun to address the relationship between narrative and argumentation more directly. (Falk and Lapesa, 2023), for example, show that stories and personal experiences can be studied within argumentative texts by annotating both narrative properties and argumentative functions. Nabhani et al. (2026) further support the view that narrativity should be treated as scalar and multidimensional rather than as a simple story/non-story difference. These studies are important because they show that narrative can play an active role in argumentative discourse. However, there is still limited work comparing how explicit and implicit argumentative components are identified across highly narrative moral stories and public argumentative talks.

Large Language Models are increasingly used to interpret, analyze, and generate natural language. They are therefore relevant for studying whether argumentative structures can be identified computationally. However, in this thesis, LLMs are not treated as the central theoretical subject. Rather, they are used as a method for investigating a broader Humanities problem: how argumentative meaning is expressed through narrative, and how difficult this meaning is to identify for both humans and computational models.

This study addresses this problem by comparing human annotations with annotations generated by three instruction-tuned LLMs. The comparison makes it possible to examine how closely LLM outputs match human annotations and what kinds of errors models make when dealing with explicit argumentation, implicit argumentative meaning, and narrative-based reasoning. In this way, the thesis combines human annotation, computational evaluation, and qualitative error analysis to study both model performance and the interpretive difficulty of argument identification across discourse types.

1.1 RESEARCH QUESTIONS AND OBJECTIVES

This thesis addresses the gap between research on narrative discourse, argument mining, and LLM-based text analysis by investigating how narrativity and discourse type influence the identification of argumentative components. The central research question guiding this study is:

How do narrativity and discourse type influence the identification of explicit and implicit argumentative components by humans and Large Language Models?

To answer this question, the study is structured around the following sub-questions:

1. How are major claims, claims, and premises expressed explicitly or implicitly in STORAL stories and TED talks?
2. What role does narrativity play in the construction of argumentative meaning in each discourse type?
3. To what extent do human annotators agree when identifying explicit and implicit argumentative components?

4. How closely do LLM-generated annotations match human annotations?
5. What kinds of errors do LLMs make when identifying argumentative components in highly narrative texts compared with public argumentative discourse containing narrative elements?

In order to investigate these questions, the study compares two datasets representing different discourse types and different roles of narrativity. The first dataset consists of folktales from the STORAL dataset, which are highly narrative and typically convey moral or social meaning through story events and character actions (Guan et al., 2022). The second dataset consists of a set of TED talk transcripts, which represent real-world public argumentative discourse where claims and supporting reasons are often more explicit, but where narrative elements still contribute to the structure and presentation of the argument.

A subset of texts from both datasets is manually annotated for argumentative components, including major claims, claims, and premises. The annotation framework also distinguishes between explicit and implicit components, which makes it possible to examine how argumentative meaning is expressed differently across the two corpora. The human annotations are then compared with LLM-generated annotations in order to evaluate model performance and identify recurring error patterns.

By combining insights from argumentation theory, narrative theory, and computational analysis, this thesis contributes to a better understanding of how argumentative meaning is expressed and identified across discourse types. More specifically, it examines how narrativity affects both human and computational interpretation of argumentative components. In doing so, the thesis contributes to ongoing discussions in Digital Humanities regarding the use of computational methods to analyze complex forms of meaning in cultural texts.

2 | LITERATURE REVIEW

2.1 THEORETICAL BACKGROUND

This study is situated at the intersection of argumentation theory, narrative theory, and computational approaches to discourse analysis. In order to investigate how Large Language Models (LLMs) extract argumentative components across different discourse types, it is necessary to define the key concepts underlying this research, including narrativity, argumentative structures, and the relationship between narrative and argumentation.

This literature review therefore first discusses narrativity as a concept, then defines argumentative structures as used in argument mining, and then examines how narrative and argumentation can overlap. Finally, it reviews computational work on argument extraction, moral and narrative datasets, and recent work on narratives in argumentative discourse.

2.2 NARRATIVITY AND NARRATIVE DISCOURSE

Narrativity refers to the degree to which a text conveys meaning through story elements such as characters, events, and causal or temporal progression (Üzelgün et al., 2017). Narrative texts typically present situations in which actions lead to consequences, allowing readers to infer meaning through the unfolding of events rather than through explicit explanation (Guan et al., 2022). Research in narrative persuasion has shown that such texts can influence beliefs and attitudes by immersing readers in a narrative world, a process often referred to as "transportation" (Mazzocco and Green, 2011). Through this process, readers engage with characters and scenarios, which can lead to the internalization of implicit messages and values conveyed by the story.

However, narrativity should not be treated as a simple binary distinction between narrative and non-narrative texts. Pianzola (2018) argues that narrative should be understood as a complex system, where narrative properties are interconnected and cannot be studied in isolation. According to this view, narrativity is not merely the presence of a sequence of events, but an emergent property produced by the interaction between discourse, audience and context. A key mechanism within this system is the Proteus Principle, which theorizes a many-to-many correlation between forms and functions. This principle is particularly useful for the present study because it allows narrativity to be understood as a matter of degree and functional variation, rather than as a feature that is simply present or absent.

This perspective is important for comparing STORAL stories and TED talks. STORAL texts are highly narrative because the entire text is organized as a story, where meaning is communicated through character actions and plot consequences (Guan et al., 2022). TED talks, on the other hand, are not purely narrative, but neither are they non-narrative. Instead, they usually include narrative forms, such as anecdotes, personal experiences, or background stories, which, through the Proteus Principle, are adapted to serve specific rhetorical and argumentative functions. Therefore, the contrast in this thesis is not between narrative and non-narrative discourse, but be-

tween texts in which narrativity plays different roles and is accomplished through different organizational structures.

Pianzola (2018)'s approach helps explain why narrativity is fundamentally relevant to interpretation. If narrativity emerges from the relationship between discourse, audience, and context, then identifying narrative meaning is partly an interpretive act performed by an observer. This means that annotators do not simply locate fixed narrative units, they also judge how events, temporal progression, and story-like elements functionally contribute to the meaning of the text. For the current thesis, this is especially relevant because the identification of implicit argumentative components in narrative texts depends on the annotator's interpretation of how specific narrative forms support or imply a broader claim.

For this thesis, the most important factor to be kept in mind is that narrativity should be operationalized carefully. STORAL stories can be considered highly narrative at the document level, while TED talks may contain narrative segments with specific rhetorical functions. This difference supports the two-layer annotation approach used in the study: one layer captures argumentative components, while the other captures the role of narrative organization and rhetorical function.

2.3 ARGUMENTATIVE STRUCTURES: MAJOR CLAIMS, CLAIMS AND PREMISES

Argumentation theory provides a framework for understanding how reasons support conclusions in discourse. In computational approaches to argument mining, arguments are typically analyzed in terms of components such as claims and premises. A claim is generally defined as the main standpoint or conclusion of a text, while a premise is a statement that provides a reason supporting or justifying that claim (Stab and Gurevych, 2017). This framework has been widely applied to texts in which arguments are explicitly expressed, such as persuasive essays, where claims and premises can be clearly identified and annotated (Stab and Gurevych, 2014b,a).

However, this model assumes that argumentative structures are expressed through explicit reasoning. In many real-world contexts, particularly in narrative discourse, this is not always the case. Instead, the relationship between claims and supporting reasons may be implicit, requiring interpretation rather than direct identification. Üzelgün et al. (2017) note that stories often function as "argument by another means," where the "moral" or conclusion may be left to the audience to infer rather than being stated as an explicit claim.

2.4 NARRATIVE AND ARGUMENTATION

Traditional approaches often treat narrative and argumentation as separate modes of discourse: narrative as storytelling, and argumentation as reasoning. However, this distinction is too strict for many real-world texts. Üzelgün et al. (2017) argue that narrative and argumentative perspectives often overlap in everyday discourse, and that "what is understood as narrative and argumentative" is often mixed. Their work is useful for this thesis because it provides a framework for understanding narrative and argumentation as distinct but interacting modes of discourse.

The connection between narrative and argumentation depends partly on how argument is defined. If argument is defined narrowly as an explicit relation between a premise and a conclusion, then narrative may seem external to argumentation. Stories therefore may be treated as vivid examples or emotional additions, but not as

part of the argument itself. Üzelgün et al. (2017) explain that a broader definition is needed, because argument is also contextual, interactional, and dependent on audience, values, and situation. This broader understanding makes room for narratives to contribute to argumentative meaning.

Tindale (2017) makes a similar point by arguing for a dynamic understanding of argument. Instead of treating arguments only as static products made of propositions, he argues that arguments should be understood in relation to the argumentative situation and the audience. This allows narrative arguments to be considered as part of argumentation, because stories can invite audiences to experience an issue, make an abstract idea concrete, and support a conclusion through lived or imagined experience.

From this perspective, narratives can be understood as conveying arguments through examples, where story events illustrate or imply more general claims. For instance, a folktale may not explicitly state a premise such as "lying leads to negative consequences" but this relationship is expressed through the events of the story, a process referred to as "bridging story plots and implied morals" (Guan et al., 2022). In this sense, narrative elements function as implicit forms of argumentative support (Üzelgün et al., 2017).

This is particularly relevant for computational analysis, as it challenges the assumption that arguments are always expressed through explicit reasoning structures (Stab and Gurevych, 2017). Instead, this implies that argumentative structures may be present in narrative texts in an implicit form, which differs significantly from the explicit structure typically assumed in argument mining research.

2.5 COMPUTATIONAL APPROACHES TO ARGUMENT EXTRACTION AND RELATED WORK

Recent research in computational linguistics has explored the intersection of argumentation and narrative understanding, but these areas are frequently treated as separate analytical problems. On one hand, Guan et al. (2022), who introduced the STORAL dataset, focus on the challenges of bridging story plots with implied morals. Their work highlights that understanding moral stories requires matching abstract concepts, capturing inter-event discourse relations, and aligning value preferences between story events and moral conclusions. Guan et al. (2022) therefore highlights how persuasive meaning in moral stories is structurally embedded rather than rhetorically foregrounded. However, while Guan et al. (2022) demonstrate that models struggle with discourse reasoning, they do not assess whether LLMs are capable of extracting formal claim-premise structures from these texts, or how this performance compares to human annotation.

A closely related line of work is Falk and Lapesa (2023) StoryARG corpus. StoryARG focuses on narratives and personal experiences in argumentative texts, and it is especially relevant because it combines narrative and argumentative annotation layers. At the argumentative level, StoryARG annotates the function of stories, including clarification, disclosure of harm, search for a solution, and establishing speaker authority. At the narrative level, it annotates whether a story has plot-like development, whether it is factual or hypothetical, and who the protagonist is. StoryARG is useful for the current thesis because it shows that narratives in argumentative contexts can be studied systematically and computationally. It also supports the idea that narrative spans can have argumentative functions, rather than being treated as irrelevant to argument structure. At the same time, StoryARG is different from this research. It focuses mainly on stories and personal experiences inside argumentative texts, while the present thesis compares two different

discourse types: highly narrative moral stories and public argumentative talks, as well as focusing specifically on the extraction of major claims, claims, and premises, including implicit components.

Nabhani et al. (2026)'s ARGUS framework is also relevant because it models narrative features in online argumentative discourse. ARGUS introduces a Change-MyView corpus annotated for story presence and six narrative features, combining textual features such as agency and event sequencing with reader-oriented features such as suspense, curiosity, and surprise. Importantly, ARGUS treats narrativity as a scalar property and shows that modeling narrativity in this way can better capture annotation variability than a simple binary story/non-story label. This is important for the present thesis because it supports the idea that narrativity should not be treated as a simple opposition between narrative and non-narrative discourse. Instead, narrativity may vary both between corpora and within individual texts. STORAL stories can be understood as highly narrative, while TED talks may contain narrative segments that play different rhetorical roles. ARGUS also provides a useful caution for the LLM part of this thesis: in their comparison, fine-tuned encoder models outperformed zero-shot LLMs in detecting narrativity and narrative features. This suggests that LLMs should not simply be assumed to handle interpretive narrative tasks reliably.

A different branch of research is presented by a different study, where instead of focusing on narrative inference, it aims to operationalize and measure persuasive language differently. Pauli et al. (2025) introduce the PERSUASIVE-PAIRS dataset in order to evaluate LLMs' capacity to enhance and reduce persuasive language in text rewrites. In order to evaluate persuasiveness, their work creates pairs of texts and LLM-generated rewrites using ordinal human annotation. Through this method, they show how persuasive language can be measured across different domains and how prompts and system setups have an impact on LLM results. This study reflects a broader trend in computational linguistics towards benchmarking persuasive capabilities of generative models. However, Pauli et al. (2025) conceptualize persuasion primarily as a stylistic or rhetorical quality that can be strengthened or weakened. Instead of focusing on the extraction of structural arguments, their evaluation emphasizes persuasive intensity. Although this operationalization is useful for comparing LLM persuasive creation, it ignores the fact that various discourse genres embed argumentation structures in different ways (e.g. persuasive essays, debates). Furthermore, it ignores works like fictional moral tales where the purpose of persuasion is subtly conveyed through narrative structure. Therefore, PERSUASIVE-PAIRS emphasizes quantifiable persuasive language amplification (Pauli et al., 2025), whereas STORAL emphasizes narrative-based moral inference (Guan et al., 2022).

Finally, work on LLMs and narrative generation, such as Tian et al. (2024), is relevant to the Digital Humanities framing because it shows the growing interest in using LLMs to analyze and generate complex narrative forms. However, the current thesis does not evaluate whether LLMs generate human-level narratives. Instead, it uses LLMs as a method for investigating how identifiable argumentative components are across discourse types. This distinction is important because it keeps the thesis focused on the Humanities problem: how argumentative meaning is expressed through narrative and how it can be interpreted by humans and computational models.

3 | DATA AND MATERIAL

3.1 OVERVIEW OF THE CORPORA

This study investigates how narrativity and discourse type influence LLMs’ ability to extract argumentative structures. The analysis is based on two datasets representing different levels of narrativity and discourse types. These datasets are suitable for a comparative study of argument extraction due to their difference in narrative structure, rethorical organization and degree of argumentative explicitness.

The STORAL dataset, created by [Guan et al. \(2022\)](#), consists of short folktales that use narrative events and character actions to express morals, lessons or social norms. Rather than being explicitly expressed, the argumentative meaning is often embedded implicitly within the story structure. Readers are expected to infer the broader moral, or argumentative conclusion from the series of events and their consequences. As a result, the relationship between the narrative events and argumentative meaning is indirect and interpretive.

The second dataset consists of TED talk transcripts that can be found on Kaggle ([Corral Jr, 2020](#)), which represent real-world persuasive discourse where arguments are often expressed more explicitly through claims, reasoning, and evidence. TED talks often combine storytelling, with examples, explanations and strong arguments. Argumentative structures such as claims and supporting reasons are typically more clearly expressed than in fictional moral narratives.

The comparison between these two corpora forms the main analytical contrast of the thesis. TED talks are a type of discourse where argumentation and narrative coexist but the argumentative intent is more explicitly stated, in contrast to STORAL, which is a strongly narrative speech with implicit persuasive meaning. This allows the study to investigate whether LLMs perform differently when arguments are explicitly stated versus narratively embedded.

3.2 COLLECTION AND SAMPLING

The selection of texts for annotation was carried out through a sampling process implemented in a Python notebook. The aim was to create two manageable but comparable subsets of texts from the STORAL and TED datasets. Rather than manually choosing the texts, the sampling process was based on transcript of story length, and used a fixed random seed to make the selection reproducible.

For the STORAL dataset, the original data was provided in three separate files corresponding to the train, validation and test splits. These files were loaded and merged into a single dataset. This merged dataset then contained 1779 stories in total, consisting of 1068 training rows, 355 validation rows, and 356 test rows. Each row included at least the story text and its original identifier, and in cases where a moral was available, this was also preserved as metadata. After merging the STORAL splits, additional metadata was created for each story. This included the original dataset ID, the source split, a combined source record key, character length, word length, moral length, and information about whether a moral was available. The story length was used as the basis for sampling. In order to avoid selecting

texts that were unusually short, or unusually long, only stories whose character length fell within one standard deviation of the mean story length were considered eligible for selection. The mean STORAL story length was 1387.17 characters, with a standard deviation of 1024.98 characters. This produced a lower eligibility bound of 362.19 characters and an upper eligibility bound of 2412.15 characters. After applying this filter, 1353 STORAL stories remained eligible for sampling. From this eligible subset, 20 stories were randomly sampled using random seed 42.

A similar procedure was used for the second dataset. The TED corpus was loaded from a CSV file containing 4005 rows. The required columns included the talk identifier, title, URL, description, duration and transcript. Rows with empty transcripts were removed, and metadata was created for each talk, including the original ID, source record key, transcript character length, transcript word length, and descriptive information about the talk. Transcript character length was also the starting point for filtering, same as with the STORAL dataset. The average length of the TED transcripts was 10,032.02 characters, with a 5490.74 character standard deviation. As a result, 2578 TED talks were still eligible due to them falling in the range of 4541.28 to 15,522.76 characters. Using the same random seed of 42, 20 TED talks were chosen.

Table 3.1: Length-based eligibility criteria for sampling.

Dataset	Length unit	Mean	SD	Lower bound	Upper bound
STORAL	Characters	1387.17	1024.98	362.19	2412.15
TED	Characters	10032.02	5490.74	4541.28	15522.76

Each text that was sampled was given a new project-specific identifier such as TED_001, TED_002 for the transcripts, and STORAL_001, STORAL_002 for the stories. For easier accessibility, the final samples were stored in CSV format, and separately, a folder was created where the text files were stored for each selected story or transcript. This sampling procedure resulted in a final annotation corpus of 40 texts: 20 STORAL stories and 20 TED talk transcripts. The final corpus was designed to be small enough for detailed human annotation, while still allowing for a meaningful comparison between the two discourse types.

3.3 ANNOTATION FRAMEWORK AND GUIDELINES

A central component of this study is the development and application of a two-layer annotation framework designed to capture both argumentative structure and narrative organization across different discourse types. The annotation scheme was mainly adapted from the argumentation framework proposed by [Stab and Gurevych \(2014a\)](#), which provides a well-established model for annotating argument components in persuasive essays, but was modified in order to take into account implicit argumentation.

The final annotation framework combines concepts from rhetorical discourse analysis, narrative persuasion and argumentation theory. The first annotation layer focuses on argument components such as claims and premises, while the second layer focuses on narrative structure and rhetorical functions. Together, these layers operationalize the different ways that persuasive meaning is expressed in both highly narrative texts and more explicitly argumentative discourse.

Unlike the original framework by [\(Stab and Gurevych, 2014a\)](#), the current annotation scheme does not annotate argumentative relations such as support and attack. Instead, the focus remains exclusively on the identification of argument compo-

nents themselves. This decision was taken since the study's main aim is to determine whether argumentative meaning can be found in texts with different degrees of narrativity. In addition, the annotation scheme introduces a distinction between explicit and implicit argument components. This extension was necessary since many argumentative structures in narrative texts are not directly stated, but rather implied through story events and character actions. Therefore the distinction between explicit and implicit argumentative meaning forms one of the main analytical aspects of the study.

3.3.1 Argumentation Layer

The argumentation layer identifies the fundamental argumentative components present within a text. Following argument mining (Stab and Gurevych, 2014a), the main component types were used: Major Claims, Claims, and Premises. The Major Claim represents the main standpoint, thesis or overarching conclusion of a text. In argumentative discourse such as TED talks, this often appears as the speaker's main message or conclusion. In narrative texts such as STORAL folktales, the major claim frequently takes the form of a moral lesson or implied conclusion understood through the events of the story. Claims represent the intermediate argumentative statements that contribute to the development of the major claim. They are typically evaluative or interpretive in nature and express a viewpoint that requires justification. Premises, in turn, provide the reasoning or evidence that support a major claim or a claim. Each argumentative component was additionally classified according to whether it was explicitly stated or implicitly conveyed. This resulted in six annotation labels shown in Table 3.1:

Table 3.2: Argumentation layer annotation labels.

Label	Description
MC-EXP	Explicit Major Claim
MC-IMP	Implicit Major Claim
C-EXP	Explicit Claim
C-IMP	Implicit Claim
P-EXP	Explicit Premise
P-IMP	Implicit Premise

This distinction between explicit and implicit components was especially important for the STORAL dataset, where argumentative meaning is often embedded within the narrative events, rather than expressed through direct statements. In those cases, annotators were required to infer argumentative meaning from a broader narrative context and create an explicit formulation. As a result, the annotation guidelines set a number of limitations on implicit annotation. A implicit component was only annotated when both the inferred meaning was heavily backed up by the text, and it was necessary for understanding the argumentative message. Annotators were instructed to not create multiple interpretations and to not annotate implicit components that just repeated the same idea as the major claim, just in different words.

For the explicit argumentative components, annotators were told to select the minimal textual span that represented a complete argumentative component. That would mean that a span could correspond to either a full sentence, or a smaller clause that expresses a complete point. In the guidelines, annotators were instructed to not annotate unnecessary surrounding text, including discourse markers or indicators, such as "for example" in the case of a premise, or "therefore", "I believe that" in the case that it preceded a claim.

Example 3.1 TED_001: [...] we strongly believe that **[by reimagining the role and the agency of the architect, we have set up the core idea**

of the project into motion; that is, to transform the river from sewage to public space for all, thereby making sure that the city of Fez will remain a living city for its inhabitants rather than a mummified heritage][MC-EXP]

Furthermore, there was a strict rule added regarding punctuation. Punctuations at the end of an argument component should not be included in the annotation, unless it is included in a quote or a direct speech, in order to have the same structure as the output of the LLM, and therefore ensuring small details such as punctuation do not affect the evaluation results. This was especially important for the TED dataset, where direct speech and quotes are often used to support claims. Example 3.2 shows an appropriate time when the punctuation was used, because it is located within the quote, whereas in Example 3.3 the punctuation is incorrectly annotated for these specific guidelines, since the punctuation is not part of a quote, and it is located at the end of the argument component, therefore should not be in the span.

Example 3.2 TED_020: [5% said, 'My life is a play.'] [P-EXP]

Example 3.3 TED_001: Actually [the state of the river before entering the medina is pretty healthy.] [C-EXP]

An exception to the punctuation rule was made in certain STORAL annotations involving implicit argument components. The supporting span occasionally had to include multiple consecutive sentences in order to offer enough textual context for the inferred component. This was due to the fact that argumentative meaning in these tales is often spread out across various events and phrases. In these situations, preserving the coherence of the supporting evidence was considered to be more crucial than the strictly following sentence-level span boundaries. Additionally, this method prevented the unnecessary duplication of explicit formulations for a single implicit argumentative component.

Implicit argumentation components provide additional challenges since the argumentative meaning is not stated openly in the text, whereas explicit argumentative components may be immediately connected to textual spans. As a result, the annotation of implicit components required a more complex three-step process that can be seen in Example 3.4. First, annotators select a supporting span from which implicit meaning could be inferred. Second, they assigned the corresponding implicit label (MC-IMP, C-IMP, P-IMP). And third, they wrote an explicit formulation of the implicit meaning as a complete sentence. The explicit formulation was implemented in order to capture the annotator's interpretation of the unspecified argumentative content. Annotators were instructed to write these interpretations as generalizable statements, while still remaining as close as possible to the evidence. An implicit component annotation example can be seen below:

Example 3.4 STORAL_018:

Supporting span: "Well, young man... there is only one advice I can give you. [the secret of happiness is to see all the beauty and marvels of the world but never to forget the drops of oil on the spoon], said the wise man."

Label: MC-IMP

Explicit formulation: You need to have balance between enjoying life and having responsibilities.

Furthermore, annotators were instructed to not rely on the moral statements provided in the original STORAL dataset. This requirement ensured that the annotation process captured the annotators' understanding of the story instead of retrieving data from an existing dataset. There is a key difference between the supporting span and the explicit formulation that is especially helpful for the compu-

tational evaluation. The explicit formulation contains the interpretative argumentative meaning that is inferred from the span, whereas the supporting span captures the actual textual evidence in the text that supports said inference. As a result, implicit argumentation is treated by the annotation framework as a unique interpretative occurrence that has to be explicitly operationalized rather than as missing information. The annotation guidelines also recognize that the highly narrative texts may have a very different argumentation structure compared to a more explicitly persuasive discourse. Furthermore, in shorter texts like often encountered in the STORAL dataset, the argumentative structure may just be represented through a major claim alone, or a major claim combined with a few premises, without a clear intermediate claim necessarily appearing. For that reason, these instances were not considered as annotation failure, but rather a valid outcome.

3.3.2 Narrative Layer

In addition to argumentative annotation, the study introduced a second analytical layer focusing on narrative organization and rhetorical function. The purpose of this layer was to operationalize narrativity as a structural feature that shaping the communication of persuasive meaning rather than only as a thematic characteristic of texts. This layer draws on theories of narrative persuasion and classical rhetorical structure (Purdue, 2023). Narrativity refers to the degree to which a text conveys meaning through story elements such as characters, events, and causal or temporal progression (Üzelgün et al., 2017). Narrative texts typically present situations in which actions lead to consequences, allowing readers to infer meaning through the unfolding of events rather than through explicit explanation (Guan et al., 2022).

Unlike the argumentation layer, which operates mainly at the span level, the narrative layer combines document-level classification with functional span annotation. First, texts were grouped based on their narrative structure and general discourse style. While STORAL tales were regarded as worldwide narrative texts in which narrative structure dominates, TED talks were typically categorized as argumentative or persuasive discourse with narrative elements. The narrative layer was based on categories derived from classical rhetorical theory. Narrative spans within argumentative discourse were annotated according to their rhetorical function within the larger persuasive structure. These categories included can be seen in Table 3.2.

Table 3.3: Classical rhetorical labels and descriptions.

Label	Description
Exordium	The introduction, opening, or hook.
Narratio	The context or background of the topic.
Propositio	The claim, stance, and argument.
Confirmatio & Refutatio	Positive proofs and negative proofs of support or evidence.
Peroratio	The conclusion and call to action.
Other	Text that does not fit into any of the rhetorical categories above.

The Exordium category typically includes opening anecdotes, personal stories, illustrative examples, or other narrative devices used to engage the audience's attention and introduce the topic of the discourse. It serves as an opening point into the argument and is typically found at the start of a text. Instead of making a claim straight away, the Exordium seeks to spark curiosity, engage the audience, and give a clear overview of the topic that will later be developed through argumentation.

Narratio refers to narrative segments that provide background information necessary for understanding the argument. Descriptions of events, situations, experiences, or general conditions relating to the subject at hand may be included in these spans. Narratio's main purpose is to place the argument in a larger perspective and provide the audience the knowledge they need to fully understand the importance

of the topic at hand. Before the speaker presents their stance, these parts in argumentative speech frequently describe why an issue arises or how a certain scenario came to be.

Propositio outlines the writer's or speaker's main standpoint and indicates the course of the argument. When a narrative span introduces the main idea, sets the speaker's main point of view, or establishes the line of reasoning that organizes the rest of the discourse, it is marked as Propositio. These segments function as a link between Narratio's contextual information and the argument's development in the future.

Confirmatio and Refutatio cover the narrative segments that may function as evidence by illustrating or supporting a claim. These include anecdotes, case studies, or descriptions of specific events that demonstrate a point. In some cases, narrative evidence may also implicitly challenge alternative viewpoints by presenting contrasting outcomes. Its function is to provide concrete narrative support for the argument and strengthen its persuasive force.

The argument's last phase is shown by Peroratio. When narrative spans help to conclude the discourse by summarizing important concepts, reiterating the main point, or leaving a last impression that supports the text's general argumentation goal, they are noted as Peroratio. These sections often go over topics that were covered previously in the speech and relate them to the speaker's conclusion or call to action.

Lastly, narrative components that add to the discourse but do not obviously fit into any of the above rhetorical roles, fall under the Other category. Quotations that strengthen the argument, narrative passages, or stylistic narrative characteristics whose rhetorical function cannot be clearly categorized as Exordium, Narratio, Propositio, Confirmatio/Refutatio, or Peroratio are appropriate examples. This category was applied only in situations when no other term could properly describe the purpose of the narrative span.

It is crucial to remember that these classifications represent the rhetorical role that narrative segments play in argumentative speech rather than the location of the narrative structure. As a result, a section is not marked as Peroratio just because it is at the end, or as Narratio just because it appears early in the text. Rather, the narrative segment's relevance to the argument's development, framing, conclusion, or support is the base of the annotation decisions. The two datasets used the narrative layer in various ways. STORAL tales were not separated into fine-grained narrative span functions since they are almost entirely narrative in nature. Rather, the texts themselves served full narrative components that discreetly conveyed meaning. TED talks, on the other hand, often included several narrative sections with various rhetorical functions inside a larger argumentative framework.

4 | METHOD

4.1 RESEARCH DESIGN

This study adopts a mixed-methods research design that combines qualitative discourse analysis with quantitative computational evaluation. The purpose of this design is to investigate how narrativity level and discourse type influence the extraction of persuasive argumentative structures by Large Language Models. The research uses an annotation-based comparison approach where computational evaluation is primarily reliant on human interpretation. The study views argument extraction as an interpretive process that involves identifying argumentative meaning, rhetorical function and implicit reasoning, rather than as a purely classification-oriented task.

Methodologically, the research combined three forms of analysis. First, the chosen texts' argumentation and narrative structures were operationalized using qualitative annotation. These annotations offered a structured representation of how human readers understood persuasive meaning. Second, the degree of similarity between human and model-generated annotations was measured using quantitative evaluation methods. Finally, in order to better understand the specific challenges that different discourse types presented for computational argument extraction, qualitative error analysis was carried out to find recurrent patterns of disagreement.

The overall design is comparative in two different ways. At corpus level, the study compares texts with significantly different levels of explicit or implicit persuasive meaning. At the annotation level, it compares multiple forms of interpretation, such as agreement between human annotation, model-generated annotations, and agreement between different annotation sources. This dual comparison enables the study to compare whether language models are able to recognize argumentative structures as well as how their performance varies in response to the text's interpretative requirements. Consequently, evaluation is not limited to only determining whether a model identifies the correct annotation label. Instead, the analysis examines multiple dimensions of annotation quality, including component identification (major claims, claims, and premises), span selection, and the reconstruction of inferred argumentative meaning.

4.2 HUMAN ANNOTATION PROCEDURE

The human annotation process formed the foundation of the evaluation framework used throughout this study. Since the objective of the research was to compare model-generated annotations against human interpretation, the creation of reliable reference annotations was one of the main goals for all analyses. As a result, the annotation process was designed to accommodate the interpretative nature of argumentative and narrative discourse annotation, while striving for consistency and transparency. The annotation was therefore conducted by two annotators with prior knowledge of argumentation theory and discourse analysis, who followed the guidelines that were described in depth in Chapter 3. Before beginning the full process, both annotators familiarized themselves with the guidelines and discussed the definitions of

the annotation categories in order to establish a shared understanding of the framework.

4.2.1 Annotation Environment

All annotations were performed using the INCEpTION annotation platform (Klie et al., 2018). INCEpTION was selected because it supports span-based annotation, custom annotation schemas, and multilayer annotation projects, making it particularly suitable for the requirements of this study. The platform allowed annotations to select textual spans directly within the source documents and assign pre-defined labels according to the annotation scheme. Rather than treating implicit formulations as a separate annotation feature, implicit and explicit argumentative components were incorporated directly into the label inventory. Consequently, annotations selected the appropriate label depending on whether the argumentative meaning was explicitly expressed in the text or inferred from contextual evidence.

4.2.2 Annotation Procedure

The annotation process was conducted independently by both annotators. For each text, annotators first read the complete document in order to gain an overall understanding of its argumentative and narrative structure. This preliminary reading was particularly important because many annotation decisions depended on contextual interpretation rather than isolated sentence-level analysis. Following the initial reading, annotators identified the potential argumentative and narrative spans and labeled them in accordance with the annotation rules. In order to annotate explicit components, the textual span that explicitly expressed the component was chosen, and then its appropriate label was assigned. Additionally, annotators created an explicit formulation that reflected the inferred argumentative meaning for implicit components. These formulations played a crucial role in the annotation process as they were meant to convey the interpretation that underlined the annotation. Throughout this process, annotators were instructed to avoid over-interpretation and to ground all annotation decisions in textual evidence. In cases involving implicit argumentation, inferred meanings were required to be supported by the broader context of the text, rather than by isolated sentences.

4.2.3 Guideline Refinement

The annotation process involved multiple rounds of refinement. The first annotation round was conducted on the TED corpus, consisting of twenty TED talk transcripts. The decision to begin with TED talks was made deliberately. Compared to STORAL stories, TED talks generally are made up of a larger amount of explicit argumentative structures, making them a more suitable starting point in order for the annotators to familiarize themselves with the annotation framework and the distinction between argumentative component types. After the first round, agreement scores were calculated in order to assess the consistency of the annotations. The results revealed several sources of disagreement, such as span boundaries, differentiation between claims and premises, and inconsistencies in the assignment of rhetorical narrative functions. While the annotation guidelines provided definitions for each category, the actual application of these definitions occasionally led to different interpretations. Therefore, based on the observed disagreements, the annotation guidelines were revised and clarified. Following those revisions, a second annotation round was conducted in order to improve annotation consistency and ensure that the two annotators followed the rules similarly.

4.2.4 Annotation Disagreements

The agreement improvement process focused specifically on texts that scores lower levels of annotation agreement. Texts with Gamma agreement score below 0.50 were selected for review and discussion. More about the Inter-Annotator Agreement Gamma on the Section 4.3. Rather than reannotating the entire corpus again, the annotationators focused only on those texts where disagreements occurred. The review process began with a collaborative evaluation of one text with low score in order to identify the most common sources of disagreement. Annotators compared their annotations span by span and analyzed examples with contradictory labels, different interpretations, or significantly differing span bounds. When one annotator agreed that the alternative annotation represented the guidelines more accurately, then the annotation was changed accordingly. In situations where span differences meant only making small changes that did not alter the meaning of the component, then annotations were often left unchanged. For disagreements that involved fundamentally different interpretations, such instances were documented and discussed further before a final decision was reached. This procedure served two purposes. First, it improved consistency between annotators by encouraging a shared understanding of the guidelines. Second, it offered insightful feedback regarding the uncertainties of the guidelines, allowing further improvements to be made as needed.

4.2.5 STORAL Gold Annotation

For the STORAL corpus, a different strategy was adopted. Whereas the TED corpus primarily served as a foundation for measuring Inter-Annotator Agreement, the annotation of STORAL revealed a methodological challenge concerning the implicit argumentation. Initially, both annotators annotated separately the STORAL stories, using the same guidelines used for the TED corpus. Afterwards, each story was reviewed collectively, where the explicit formulations related to implicit argument components received special attention. Because the STORAL stories communicate persuasive meaning through implied morals and narrative inference, annotators often produced explicit formulations that differed in wording, but expressed very similar interpretations. Therefore, during these conversations, the annotators then reviewed the textual evidence supporting each formulation, compared their interpretations, and reached a consensus on a final version. As a result of this process, each STORAL story was given a single gold-standard annotation, which included a single agreed-upon explicit formulation for each implicit argument component. The creation of a consensus gold annotation served an important methodological function. Since the evaluation of implicit argumentation depends on both span identification and the reconstruction of inferred meaning, establishing a consistent reference decreased the variation resulting from different but equally plausible formulations. Consequently, instead of adding more complexity to the evaluation process, the resulting annotation offered a more reliable standard for the subsequent evaluation of Large Language Model outputs.

The final human annotation dataset consisted of layered argumentative and narrative annotations produced through independent annotation, iterative guideline refinement, disagreement resolution, and consensus building.

4.3 INTER-ANNOTATOR AGREEMENT

Following the human annotation procedure, Inter-Annotator Agreement (IAA) was calculated in order to assess the consistency of the annotations produced by the two annotators. This step was necessary because the human annotations later served as

the reference point for evaluating the outputs generated by Large Language Models. Before using the annotations as reference data, it was important to examine the extent to which the annotators applied the annotation guidelines in a comparable way.

Agreement was calculated using the Gamma agreement measure which was originally proposed by [Mathet et al. \(2015\)](#) as a unified method for measuring agreement over sequences of annotations. Gamma was selected because the annotations in this study are span-based rather than based on fixed predefined units. In other words, annotators were not asked to assign labels to a fixed list of sentences or tokens. Instead, they independently selected text spans and assigned labels to those spans. This means that disagreement could occur at multiple levels: annotators could disagree about whether a passage should be annotated at all, where the beginning and end of a span should be placed, or which label should be assigned to a selected span.

According to [James \(2026\)](#), traditional inter-annotator agreement measures such as Cohen's Kappa are less suitable for this type of task because they assume that annotators classify the same predefined units. This was not the case in the present annotation task. For example, two annotators might both identify the same general argumentative component, but one annotator might select a full sentence while the other selects only the central clause. Similarly, two annotators might select overlapping spans but assign different labels, such as Claim and Premise. Gamma is therefore more appropriate because it is designed for annotation tasks involving both segmentation and categorization ([James, 2026](#)).

In this thesis, Gamma was computed using the Python package `pygamma-agreement`, developed by [Titeux and Riad \(2021\)](#). The package provides a programmable Python implementation of Gamma, making it suitable for integration into the notebook-based evaluation pipeline used in this study. The use of a Python implementation was important because the annotation files had to be processed, filtered, and evaluated across multiple documents and annotation layers.

The agreement calculation was implemented in a Jupyter notebook. The notebook first installed the required packages, including `pygamma-agreement` and the optional CBC solver extension. The analysis then imported the relevant Python libraries, including `json`, `os`, `pandas`, `numpy`, `Continuum` from `pygamma-agreement`, `Segment` from `pyannote.core`, and `CombinedCategoricalDissimilarity` from `pygamma-agreement`. These libraries were used to read the annotation files, convert span offsets into comparable annotation units, define label dissimilarity settings, compute Gamma, and save the resulting scores.

A custom parsing function was written to extract the relevant annotation information from the INCEpTION JSON files. The parser read through the structure of each feature and identified annotation objects belonging to the custom annotation layers used in the project. These layers corresponded to the argumentation and narration annotation layers. For each annotation, the parser extracted the start offset, end offset, annotator name, and assigned label. Because INCEpTION stores internal feature names separately from the readable label names shown in the user interface, the parser also created a mapping between internal feature names and their corresponding UI names. This ensured that labels such as implicit and explicit argument components could be represented using the readable labels used throughout the thesis, such as MC-EXP, C-IMP, or P-EXP. This mapping step was important because the agreement analysis needed to compare the annotation categories themselves, not the internal encoding used by the annotation platform. The parser also excluded helper text fields associated with implicit formulations, such as `IClaimtext`, `MCIMPtext`, and `PIMPtext`. These fields were ignored because Gamma was used to evaluate span-based labels rather than the content of the written explicit formu-

lations. The explicit formulations are relevant for later semantic comparison, but they do not function as span categories in the Gamma calculation. Therefore, only the span labels themselves were included in the inter-annotator agreement analysis. These labels were then divided into two groups which corresponded to the two annotation layers.

The dissimilarity between annotation categories was calculated. This made it possible to specify different values for α and β , which control how segmentation and categorization differences are penalized. Alpha accounts for the positional discrepancies, meaning differences in unit boundaries, and Beta accounts for the categorical discrepancies, meaning the differences in labels (James, 2026).

Several Gamma configurations were calculated for each document. First, a standard Gamma score was computed using $\alpha = 1$ and $\beta = 1$. This configuration treated span-boundary disagreement and label disagreement with equal weight. Second, a soft-Gamma was computed using the same parameter boundaries. The soft version was created by Titeux and Riad (2021) with the goal of doing exactly what gamma does, except it reduced the disagreement caused by splits in annotations (Riad et al., 2020), therefore it is more tolerant to boundary differences between overlapping annotations. This was useful for this research because small span-boundary differences were common, especially when annotators identified the same component, but selected slightly different textual boundaries. In addition to the equal-weight configuration, a second configuration was calculated with $\alpha = 1$ and $\beta = 2$. This setting placed a stronger penalty on label disagreement. The purpose of this additional configuration was to test how agreement changed when category mismatch was treated as more serious than boundary mismatch. This was important because some disagreements in the annotation data could have involved overlapping or nearly overlapping spans with different labels. By comparing the results of the two configurations, the analysis could distinguish between disagreement caused mainly by span-boundary variation and disagreement caused by different interpretations of the annotation category. For each TED document, the notebook calculated twelve Gamma scores: four for all annotations combined, four for the argumentation layer only, and four for the narration layer, all which included the standard and soft Gamma with both parameter settings.

The narrative layer needed some special handling, due to the fact that in some cases, one annotator did not produce any narrative annotations for the document, which would make the direct comparison difficult. To avoid excluding these cases entirely, the notebook included an option to add a dummy none span for a missing annotator in the narration-layer calculation. This allowed Gamma to still be computed in cases where one annotator had no annotations in the selected layer. This step was applied only where necessary and only for the purpose of enabling layer-specific agreement calculation.

4.4 MODEL SELECTION AND PROMPT OPTIMIZATION

After the human annotation and agreement analysis stages, the next step of the study involved generating annotations with Large Language Models. The goal of this stage was to test whether instruction-tuned LLMs could apply the annotation framework described in Chapter 3 to the same TED talk and STORAL texts annotated by humans. The models were not fine-tuned on the annotation task. Instead they were used in an inference-only setting, meaning that the models were used solely to generate outputs based on their existing pre-trained knowledge without updating or training their parameters. In this setup, the complete task definition, label inventory, annotation rules, examples, and output format were provided through prompts, and the models produced annotations directly from these instructions.

The LLM annotation stage was conducted using three instruction-tuned open weight models from different families. The decision to use multiple models was motivated by the fact that prompt-based performance can vary significantly across different model families. Previous work on prompt optimization has shown that the effectiveness of a prompt is not only dependent on the task, but also on the specific model to which the prompt is applied. In other words, a prompt that works well for one model, may not necessarily produce the best output for another. For this reason, prompt development in this study is treated as both a task-specific and model-specific process (Chen et al., 2023).

All models were accessed through a Python prompting pipeline using an OpenAI-compatible client. The prompts were sent as a system message and a user message. The system message defined the role of the model as an expert annotator and specified high-level output constraints, while the user message contained the full annotation task, definitions, rules, examples where applicable, and the text to be annotated. For all final prompting runs, the temperature was set to 0 in order to make the generations as deterministic as possible.

4.4.1 Model Selection

The models were selected for three main reasons. First, all three models are instruction-tuned, meaning they are designed to follow natural-language instructions and produce task-oriented outputs. This was important because the model were not fine-tuned for the annotation task, instead they were expected to perform the annotation through prompting alone. Second, the models belong to different model families, which made it possible to examine whether different instruction-tuned LLMs behave differently when asked to apply the same annotation guidelines. Third, the selected models are relatively comparable in scale, ranging from approximately 24B to 31B parameters. This allowed the comparison to focus more on model behavior and instruction-following ability rather than on extreme differences in model size.

The first model used was google/gemma-4-31b-it. This model was included because it represents the Gemma model family and provides a large instruction-following model suitable for complex text analysis tasks. Its size made it a relevant candidate for annotation tasks requiring discourse-level interpretation, particularly because the prompts used in this thesis were relatively long and included definitions, rules, examples, and output constraints.

The second model was mistralai/mistral-small-3.2-24b-instruct. This model was selected because it is an instruction-tuned Mistral model explicitly designed for improved instruction following. This was relevant to the present study because the task required the model to follow several constraints at once: identify only relevant spans, copy spans exactly from the text, assign labels from a fixed inventory, avoid unsupported interpretation, and return the output in a structured format. The model was therefore suitable for testing whether a compact but capable instruction model could perform complex annotation according to detailed guidelines.

The third model used was qwen/qwen3.6-27b. This model was selected in order to include a model from the Qwen family and to compare its behavior with the Gemma and Mistral models. Like the other models, Qwen3.6-27B was used in an instruction-following setting rather than fine-tuned on the annotation task. Its inclusion allowed the study to compare whether a model from a different development family would show different strengths or weaknesses in argument extraction, implicit formulation, and adherence to structured output requirements.

4.4.2 Prompt Development and Optimization

The use of prompt-based annotation rather than supervised fine-tuning was a deliberate methodological choice. The aim of the study was not to train a specialized argument mining model, but to evaluate whether contemporary instruction-tuned LLMs can apply complex annotation guidelines in a zero-shot or few-shot setting. This reflects a realistic use case in which LLMs are used as annotation assistants without task-specific training data. In-context learning (ICL) is particularly relevant in this setting because it allows a model to perform a task using instructions and examples supplied directly in the prompt, while the model parameters remain unchanged (Li, 2023).

Prompt development was one of the most important stages of the computational methodology. The initial prompts were designed as zero-shot prompts. In the first versions, the models were provided only with a short task description, the annotation labels, and the text to annotate, without any examples of what should be annotated, and with minimal to no details about the definitions of the labels. This initial stage was used to observe how the models responded to a relatively simple annotation task and to establish a basic point of comparison for later prompt versions.

As prompt testing progressed, the amount of information included in the prompts was gradually increased. Later versions contained longer and more precise definitions of each annotation label, more detailed span-selection rules, and clearer restrictions on what should and should not be annotated. In the final versions, the label definitions were aligned as closely as possible with the definitions used in the human annotation guidelines. This was done so that the LLMs and the human annotators were, as close as possible to working from the same conceptual description of the annotation task. The development process therefore moved from broad task instructions toward prompts that closely resembled the human annotation guidelines in both terminology and level of detail.

Few-shot examples were introduced in a second phase of prompt development. These examples demonstrated to the model what an appropriate annotation should look like in terms of output formatting and span selection. Because implicit components needed two distinct operations, choosing the textual evidence and developing the inferred argumentative meaning, few-shot prompting was particularly helpful for implicit argumentation. Examples were given on how to annotate the supporting spans, how to assign the label and when to provide an explicit formulation. But using examples also had to be done with caution. Few-shot examples can improve performance in many contexts, according to earlier research done by Li (2023) and Chen et al. (2025), but they can also add bias or cause the model to mimic the examples too closely rather than applying the whole instructions to the new text. Because of this, the few-shot examples were chosen to illustrate the reasoning style and annotation format without pushing the model to draw too many conclusions from a single instance. In order to demonstrate situations in which no annotation should be created, negative examples were also provided. This type of examples included non-argumentative material such as greetings, factual or descriptive statements that do not provide an argumentative or narrative function.

Prompt optimization was carried out on a development subset which consisted of 20% of the full dataset, therefore being made up of four TED talk transcripts and four STORAL stories. The development subset was used to test different prompt versions before applying the final prompts to the remaining sixteen texts from each dataset. The purpose of this step was to avoid designing prompts that were too general. Instead, prompts were evaluated on actual texts from both discourse types so that their performance could be assessed on both explicitly persuasive discourse and highly narrative moral stories. The TED and STORAL prompts required differ-

ent forms of optimization. For the TED corpus, the prompt focused mainly on identifying explicit and implicit argumentative components in a persuasive discourse context. TED talks often contain direct claims, supporting examples, personal anecdotes, and rhetorical transitions. Therefore, the prompt had to instruct the model to distinguish between argumentative content and surrounding rhetorical material. For the STORAL corpus, the prompt focused on implicit argumentation. Since STORAL stories often communicate persuasive meaning through events and implied morals rather than through direct claims, the prompt instructed the model to identify supporting spans and generate explicit formulations of the inferred argumentative meaning.

The prompt optimization process was also model-specific. The same general annotation task was given to all three models, but prompt variants were adjusted when a model repeatedly failed to follow a particular requirement. For example, if a model returned invalid JSON, later prompt versions emphasized the output format more strongly. If a model produced overly broad spans, the prompt stressed minimal span selection. If a model paraphrased the source span rather than copying it exactly, the prompt was modified to explicitly prohibit paraphrasing. In this sense, prompt optimization was not treated as a single universal solution, but as an iterative process of adapting instructions to both the task and the behavior of each model. This approach is supported by research on model-adaptive prompt optimization, which argues that different LLMs may require differently optimized prompts even for the same downstream task (Chen et al., 2023).

Several principles guided the final prompt design. First, the prompts used explicit task instructions. The model was told exactly what type of annotation to perform and which labels were allowed. Second, the prompts used structured definitions. Each label was defined in relation to the annotation guidelines used by the human annotators. Third, the prompts included output constraints, especially the requirement to return structured JSON. Fourth, the prompts included negative restrictions, such as avoiding unsupported inferences, avoiding paraphrased spans, and avoiding additional explanatory comments outside the required output. Fifth, the prompt provided many examples to steer the model towards the desired performance.

The final prompts therefore combined zero-shot instruction, few-shot demonstration, and explicit formatting constraints. They were not simply questions asking the models to "find arguments" in the text. Instead, they functioned as detailed annotation instructions that translated the human annotation guidelines into a form that could be applied by LLMs. The final prompt versions were then used to generate model annotations for the full evaluation dataset. All final prompts are included in Appendix A in order to make the computational annotation procedure reproducible.

4.5 LLM EVALUATION

After the LLM-generated annotations were produced, the outputs were evaluated against human reference annotations. Because the TED talk and STORAL tasks required different types of output, two evaluation strategies were used. For the TED corpus, the evaluation focused on span-based annotation matching, since both the human annotators and the LLMs selected text spans and assigned labels to them. For the STORAL corpus, the evaluation focused more strongly on semantic similarity, since the task required the models to formulate implicit argumentative meaning in addition to selecting supporting spans. For the TED corpus, model outputs were evaluated using precision, recall, and F1. Precision measures the proportion of model-produced annotations that matched the human reference annotations, while

recall measures the proportion of human reference annotations that were recovered by the model. F1 was used as the harmonic mean of precision and recall:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

This metric was used because the task is an extraction task, in which the main concern is whether the model identified the relevant annotated units rather than whether it correctly identified the large amount of non-annotated text. Precision, recall, and F-measure are commonly used in information retrieval and classification evaluation (Powers, 2020), and F1-based evaluation is also common in argument mining tasks involving the identification of argumentative components (Stab and Gurevych, 2017).

Several F1-based variants were calculated to account for different degrees of span-boundary strictness. First, a span-level evaluation compared predicted and reference spans based on their overlap, allowing partial matches to contribute to precision and recall, similar to approaches used in span-identification tasks such as SemEval-2020 Task 11 (Da San Martino et al., 2020). A stricter variant applied an intersection-over-union threshold, where predicted spans were only counted as correct if they sufficiently overlapped with a reference span of the same label. In addition, a character-offset evaluation measured overlap at a more fine-grained level, by comparing the character positions covered by predicted and reference annotations. Finally, a sentence-level evaluation mapped spans to sentences and compared the selected sentence units, providing a less boundary-sensitive perspective when models identified relevant content but differed in exact span boundaries.

For the TED corpus, evaluations were performed separately for each model and each human annotator. This allowed the model outputs to be compared against both sets of human annotations rather than assuming that one annotator’s annotations alone represented the only possible reference. The same evaluation code was used for the argumentation and narrative layers, with the relevant label set changed according to the annotation task.

The STORAL evaluation required a different approach. Unlike the TED argumentation task, the STORAL task focused on implicit argumentative components. For each implicit component, the model had to provide a label, a supporting span, and an explicit formulation of the inferred argumentative meaning. Since the explicit formulation is a generated sentence rather than an extracted span, exact textual matching would not be appropriate as the main evaluation method. Two semantic similarity metrics were therefore used: SBERT cosine similarity and BERTScore.

SBERT was used to compare the human and model-generated explicit formulations. SBERT was introduced by Reimers and Gurevych (2019) with the aim to produce sentence embeddings that are designed to be semantically meaningful and comparable using cosine similarity. In this study, the sentence-transformers model all-mpnet-base-v2 was used to encode the human explicit formulation and the LLM explicit formulation. The cosine similarity between the two embeddings was then calculated. Higher scores indicate stronger semantic similarity between the human interpretation and the model-generated interpretation. BERTScore was used as a second semantic evaluation metric. Introduced by Zhang et al. (2019), BERTScore compares a candidate text with a reference text using contextual token embeddings rather than relying only on exact word overlap. In this study, the human explicit formulation was treated as the reference and the LLM explicit formulation was treated as the candidate. BERTScore precision, recall, and F1 were calculated using roberta-large with English language settings and baseline rescaling. This provided a second way to estimate whether the model-generated implicit formulation was semantically close to the human formulation. For both SBERT and BERTScore,

annotations were compared within the same label category. Human and model annotations were first grouped by label: MC-IMP, C-IMP, and P-IMP. When both the human reference and the model output contained annotations with the same label, a similarity matrix was created between all human and model formulations for that label. The best pairing between human and model annotations was then selected using an assignment procedure that maximized the overall similarity score. This ensured that the human annotation and each model annotation could be matched at most once.

In addition to semantic similarity between explicit formulations, the STORAL evaluation also compared the supporting spans selected by the human reference and by the model. This was done using token-overlap F1. The supporting spans were lowercased and tokenized, and precision, recall, and F1 were calculated based on the overlap between the token sets. This supporting-span score was included because the task did not only require the model to formulate an implicit meaning, but also to identify the textual evidence from which that meaning was inferred. If a human annotation had no corresponding model annotation with the same label, it was recorded as missed by the LLM. If the model produced an annotation for which there was no corresponding human annotation with the same label, it was recorded as an extra LLM annotation. These cases were preserved in the detailed evaluation output so that they could later be inspected during the qualitative error analysis.

4.6 COMPARATIVE ANALYSIS

In addition to the F1-based evaluation described above, a comparative Gamma analysis was conducted for the TED corpus. While F1 was used to evaluate model outputs against human reference annotations, Gamma was used to compare complete annotation sets across different annotation sources. This allowed the study to examine not only model-human similarity, but also the degree of agreement between individual humans and individual models, and between the models themselves. This comparative Gamma analysis was limited to the sixteen TED talks for which the final LLM outputs were available. The four TED talks used during the prompt development were therefore not included. The same procedure was applied to both the argumentation layer and the narrative layer, using the respective labels of each. In both cases, the purpose was to compare span-based annotations produced by different sources using the same agreement framework.

The analysis used the pygamma-agreement implementation of the Gamma measure, following the procedure described earlier for inter-annotator agreement (Mathet et al., 2015; Titeux and Riad, 2021). Gamma was appropriate for this stage because the annotation task involved both segmentation and categorization. Different annotators or models could disagree not only about the label assigned to a span, but also about where the span began and ended, or whether a span should be annotated at all. Before Gamma could be calculated, all annotations were converted into a shared format. Human annotations were extracted from the INCEpTION JSON exports. The script parsed the annotation layers, retrieved the start and end offsets of each span, and mapped internal INCEpTION feature names to their readable annotation labels. LLM annotations were read from the model-generated JSON files. Since the LLMs produced copied text spans rather than INCEpTION offsets, each model span was located in the original TED transcript and converted into character start and end offsets. This made the human and model annotations comparable within the same text continuum. The final model outputs were loaded for our three models: Mistral-Small-3.2-24B-Instruct, Qwen3.6-27B, and Gemma 4 31B. For each document, the script loaded the relevant human annotations and the corresponding LLM annotation files. The model output filenames encoded the document ID,

model name, prompt version, and task type, allowing each annotation file to be linked to the correct TED transcript and annotation layer.

The comparative Gamma analysis was organized into four groups. First, human-human Gamma was calculated by comparing Annotator_1's annotations with Annotator_2's annotations. This provided the human agreement baseline for the same set of sixteen final TED documents. Second, model-human Gamma was calculated by comparing each model separately with each human annotator. This produced separate comparisons for Mistral versus Annotator_1, Mistral versus Annotator_2, Qwen versus Annotator_1, Qwen versus Annotator_2, Gemma versus Annotator_1, and Gemma versus Annotator_2. Third, pairwise model-model Gamma was calculated for each pair of LLMs. This included Mistral versus Qwen, Mistral versus Gemma, and Qwen versus Gemma. Fourth, a multi-source model-model Gamma calculation was conducted across all three models together. This final comparison measured agreement among Mistral, Qwen, and Gemma as a group.

For each comparison, the same four Gamma variants were calculated: standard Gamma with $\alpha = 1$ and $\beta = 1$, soft Gamma with $\alpha = 1$ and $\beta = 1$, standard Gamma with $\alpha = 1$ and $\beta = 2$, and soft Gamma with $\alpha = 1$ and $\beta = 2$. The standard and soft versions were included to compare stricter and more boundary-tolerant agreement settings. The two beta settings were included to test the effect of assigning a stronger penalty to category disagreement.

The script also recorded the number of annotations produced by each source in each comparison. For human-human comparisons, the output included the number of annotations produced by Annotator_1 and Annotator_2. For model-human comparisons, it included the number of annotations produced by the selected human annotator and the selected model. For model-model comparisons, it included the number of annotations produced by each model. These counts were preserved so that Gamma scores could later be interpreted in relation to the amount of annotation produced by each source. Finally, a summary table was generated by averaging Gamma scores by agreement type, task type, annotator or model combination, Gamma type, alpha value, and beta value. These outputs were used in the results chapter to compare the different sources of agreement across humans and models.

4.7 ERROR ANALYSIS

In addition to the quantitative evaluation, a qualitative error analysis was conducted in order to examine the types of disagreement that occurred between the human annotations and the LLM-generated annotations. While the quantitative metrics provided an overview of model performance, they did not fully explain why particular predictions matched or failed to match the human reference annotations. The error analysis was therefore used as a complementary method for interpreting the evaluation results.

The error analysis focused on selected examples from the TED and STORAL evaluation outputs. Cases were selected on the basis of the quantitative results, including low-scoring documents, label-level differences, missed annotations, extra model annotations, and examples where the model selected a partially overlapping but non-identical span. For STORAL, the analysis also considered cases where the model-generated explicit formulation differed semantically from the human formulation despite being based on a related supporting span.

The purpose of this analysis was not to produce a second quantitative score, but to provide a more detailed explanation of the patterns behind the evaluation results. The findings from the error analysis are presented in the Results and Discussion chapter.

4.8 IMPLEMENTATION AND REPRODUCIBILITY

All computational stages of the study were implemented in Python using a structured set of Jupyter notebooks. These notebooks cover dataset preparation and sampling, prompt execution, conversion of model outputs into evaluation-ready formats, model-human evaluation, Gamma agreement analysis, semantic-similarity evaluation for the STORAL explicit formulations, and the preparation of result tables.¹

The repository was organized according to the main stages of the project. Separate folders were used for the annotation guidelines, the TED and STORAL datasets, prompt optimization, final prompting, notebooks, and results. This organization was intended to make the workflow easier to inspect and to separate the development phase from the final evaluation phase.

The implementation relied mainly on Python libraries for data handling, text processing, evaluation, and result aggregation. In particular, pandas and NumPy were used for tabular data processing and numerical operations, while json, re, pathlib, and related standard Python utilities were used for parsing annotation files, handling file paths, and converting model outputs. The span-based TED evaluation was implemented through custom scripts for relaxed F1, overlap F1, character-offset F1, and sentence-level F1. Gamma agreement was computed in separate notebooks, while the STORAL evaluation used semantic-similarity-based comparison of explicit formulations. Overall, this setup supports transparency and reproducibility by keeping the sampled data, prompts, model outputs, evaluation scripts, and final result files in one project repository.

¹ The code, prompt files, sampled datasets, model outputs, and full result tables are available in the associated GitHub repository: https://github.com/luizatfilip10/Master_Thesis.

5

RESULTS AND DISCUSSION

5.1 GAMMA AGREEMENT ANALYSIS

This section presents the Gamma agreement results for the TED corpus. Gamma was used in two related ways: first, to measure agreement between the two human annotators, and second, to compare complete annotation sets produced by humans and LLMs. These results are presented together because the human-human comparison provides an important baseline for interpreting the model-human and model-model comparisons.

As described in the Chapter 4, Gamma is appropriate for this study because the TED annotation task was span-based. Annotators and models did not classify a fixed set of predefined units, instead, they independently selected spans and assigned labels to those spans. Disagreement could therefore occur at the level of span selection, span boundaries, annotation density, and label assignment. Gamma was selected because it is designed for this type of segmentation and categorization task (Mathet et al., 2015). The calculations were implemented using the pygamma-agreement package (Titeux and Riad, 2021).

The comparative Gamma analysis was calculated only for the final sixteen TED talks for which final LLM outputs were available. The four TED talks used during prompt development were excluded from the final model comparison. For this reason, the tables in this section report Gamma results for the same sixteen-document subset used in the final LLM evaluation. The main tables report soft Gamma with $\alpha = 1$ and $\beta = 1$. This setting was selected as the main reporting configuration because it allows some tolerance for minor span-boundary differences while still comparing annotation categories. The other Gamma configurations were used as robustness checks and are summarized separately.

It is important to note that the Gamma scores in this section are interpreted comparatively rather than through fixed universal thresholds. Because Gamma measures both how spans are selected and how they are labeled, and because this task is different from other tasks with predefined units, standard agreement categories from other measures are not directly applicable. Instead, the results are understood by comparing them to the human-human baseline and across different types of comparisons.

Table 5.1: Average soft Gamma agreement by comparison type for the final sixteen TED talks ($\alpha = 1$, $\beta = 1$).

Comparison type	Argumentation γ	Narration γ	Interpretation
Human-human	0.900	0.105	Agreement between A1 and A2. This is the human baseline for each layer.
Model-human average	0.239	0.196	Average agreement between each model and each human annotator.
Model-model pair average	0.359	0.394	Average of the three pairwise model-model comparisons.
All three models	0.384	0.429	Agreement among Mistral, Qwen, and Gemma considered together.

Table 5.1 shows a clear difference between the argumentation and narration layers. For the argumentation layer, human-human agreement was very high, with an average soft Gamma of 0.900. This suggests that, after guideline refinement and disagreement resolution, the two human annotators applied the argumentation labels in a highly consistent way on the final TED subset. In contrast, the model-human average for argumentation was much lower, at 0.239. This means that the LLM outputs were substantially less similar to the human annotations than the two human annotations were to each other.

The narration layer shows a different pattern. Human-human agreement for narration was very low, with an average soft Gamma of 0.105. This indicates that the rhetorical-function annotation of narrative spans was much less stable than the argumentation annotation. This is consistent with the qualitative difficulty of the narrative layer: labels such as Exordium, Narratio, Propositio, Confirmatio, and Peroratio are functional rather than purely positional, and the same passage can sometimes serve more than one rhetorical role. Therefore, the low human-human score should not be interpreted as a failure of one annotator, but as evidence that the narrative-function layer was more interpretive and less consistently operationalized.

Interestingly, the model-model averages were higher than the model-human averages in both layers. For argumentation, the pairwise model-model average was 0.359, compared with a model-human average of 0.239. For narration, the model-model pair average was 0.394, compared with a model-human average of 0.196. This indicates that the LLMs were often more similar to each other than they were to the human annotators. However, this should not be interpreted as evidence that the models were more correct. Instead, it suggests that the models may have shared similar annotation tendencies, such as selecting broader spans, relying on surface-level discourse cues, or applying the prompt instructions in similar ways.

Human-human agreement

The human-human comparison provides the most important baseline for interpreting the rest of the Gamma results. In the argumentation layer, the two annotators reached very high agreement. This indicates that explicit argumentative components in TED talks, such as major claims, claims, and premises, were generally identified in a consistent way. Although some disagreement remained, especially around span boundaries and claim-premise distinctions, the overall score suggests that the argumentation layer was stable enough to serve as a reference for model evaluation.

The narration layer produced much lower human-human agreement. This result is important for the interpretation of all subsequent narrative-layer model comparisons. If human annotators themselves showed low agreement on rhetorical narrative functions, then model-human disagreement in this layer cannot be interpreted in the same way as argumentation-layer disagreement. In the argumentation layer, the human reference is relatively stable, whereas in the narration layer, the human reference is itself more uncertain. Therefore, narrative-layer model results should be interpreted more carefully.

Table 5.2: Human-human Gamma agreement across Gamma configurations for the final sixteen TED talks.

Layer	Gamma type	β	Mean γ	SD
Argumentation	Standard	1	0.894	0.076
Argumentation	Soft	1	0.900	0.080
Argumentation	Standard	2	0.875	0.094
Argumentation	Soft	2	0.880	0.100
Narration	Standard	1	0.034	0.436
Narration	Soft	1	0.105	0.483
Narration	Standard	2	0.132	0.286
Narration	Soft	2	0.192	0.336

Table 5.2 shows that the difference between the two layers is evident across Gamma configurations. For argumentation, all four configurations produced high average agreement, ranging from 0.875 to 0.900. This indicates that the high human-human agreement was not dependent on a single Gamma setting. Whether standard or soft Gamma was used, and whether β was set to 1 or 2, the argumentation layer remained highly consistent.

For narration, all four configurations produced much lower scores, ranging from 0.034 to 0.192. The soft Gamma scores were somewhat higher than the standard Gamma scores, which suggests that part of the disagreement came from span-boundary or segmentation differences. However, even the soft scores remained low. This means that the disagreement was not only caused by small boundary differences, but also by broader differences in how the annotators interpreted the rhetorical function of narrative spans.

Model-human agreement

The model-human Gamma results show how closely each LLM annotation set matched the annotations of each human annotator. These comparisons were calculated separately for each model and each annotator, rather than collapsing the two human references into one. This was needed because it allowed the analysis to determine whether model agreement depended strongly on the chosen human reference.

Table 5.3: Model-human soft Gamma agreement by model and annotator for the final sixteen TED talks ($\alpha = 1$, $\beta = 1$).

Model	Arg. vs A1	Arg. vs A2	Arg. avg.	Narr. vs A1	Narr. vs A2	Narr. avg.
Gemma	0.287	0.296	0.292	0.334	0.166	0.250
Mistral	0.217	0.202	0.210	0.150	0.031	0.091
Qwen	0.214	0.216	0.215	0.308	0.189	0.248

For the argumentation layer, the model-human scores were relatively similar across the two human annotators. Gemma achieved the highest average model-human

Gamma for argumentation, with an average of 0.292. Qwen and Mistral were lower, with averages of 0.215 and 0.210. The similarity between the Annotator_1 and Annotator_2 columns is also important: for argumentation, each model obtained almost the same score against both human annotators. This suggests that the low model-human scores cannot be explained by one unusually strict or unusual human reference. Instead, the models genuinely diverged from both human annotation sets.

The contrast with the human-human argumentation score is significant. Human-human Gamma for argumentation was 0.900, while the best model-human average was 0.292. This means that the models did not come close to the human annotation behavior, even in the layer where human agreement was strong. The result suggests that the models often differed from the human annotators in what they selected, how much text they selected, or which argumentative label they assigned.

The narration layer shows a more complex pattern. Gemma and Qwen had similar average model-human scores for narration, with averages of 0.250 and 0.248 respectively, while Mistral was lower at 0.091. However, the scores differed more depending on the human annotator. For all three models, agreement with Annotator_1 was higher than agreement with Annotator_2. For example, Gemma scored 0.334 against Annotator_1 but 0.166 against Annotator_2, and Qwen scored 0.308 against Annotator_1 but 0.189 against Annotator_2. This could indicate that the measured model-human agreement varies depending on which human annotation set is used as the reference for prompting, which reflects differences between the two human annotators, rather than any direct influence of those annotations on the models themselves.

This pattern supports the interpretation that the narrative layer was less stable as a reference standard. Because the two human annotators disagreed significantly on narrative-function annotation, a model could appear more or less successful depending on which annotator it was compared against. For this reason, narrative Gamma results should be treated as exploratory and interpreted alongside the qualitative error analysis.

Model-model agreement

Model-model Gamma was calculated to examine whether the LLMs produced similar annotation patterns to each other. This comparison is useful because it shows whether disagreement with humans was model-specific or whether the models shared common annotation tendencies.

Table 5.4: Model-model soft Gamma agreement for the final sixteen TED talks ($\alpha = 1, \beta = 1$).

Comparison	Arg. mean γ	Arg. SD	Narr. mean γ	Narr. SD
Mistral vs Gemma	0.264	0.130	0.319	0.325
Mistral vs Qwen	0.293	0.157	0.443	0.330
Qwen vs Gemma	0.519	0.194	0.422	0.288
All three models	0.384	0.126	0.429	0.242

Table 5.4 shows that model-model agreement was generally higher than model-human agreement. In the argumentation layer, Qwen and Gemma had the strongest pair agreement, with a mean Gamma of 0.519. This was considerably higher than the agreement between either model and the human annotators. Mistral showed lower agreement with both Gemma and Qwen. This suggests that Qwen and Gemma may have applied the argumentation prompt in more similar ways, while Mistral produced a somewhat different annotation pattern.

In the narration layer, the model-model scores were also higher than the model-human scores. Mistral and Qwen had the highest pairwise narration agreement

at 0.443, followed closely by Qwen and Gemma at 0.422. The all-three-model Gamma was 0.429. These values suggest that the models often converged on similar rhetorical-function annotations, even though those annotations did not align strongly with the human annotators.

This model-model combination is important for interpreting the LLM results. If all models tend to select similar kinds of narrative spans or assign similar rhetorical functions, their agreement may reflect shared model behavior rather than correctness. For example, the qualitative error analysis showed that models often selected broad narrative blocks instead of the narrower rhetorical-function spans used by human annotators. If several models follow the same broad-span strategy, model-model Gamma can be relatively high even while model-human Gamma remains low.

Comparison between argumentation and narration

Taken together, the Gamma results show that argumentation and narration behaved differently as annotation tasks. The argumentation layer had a stable human-human baseline, but low model-human agreement. This indicates that the human annotators were able to apply the argumentation guidelines consistently, while the LLMs did not reproduce the same span and label decisions. In this layer, model errors are easier to interpret because the human reference is relatively reliable.

The narration layer showed low human-human agreement and low-to-moderate model-human agreement. This suggests that narrative-function annotation was not only difficult for LLMs, but also difficult for human annotators. The issue is not simply that models failed to follow a clear human standard. Rather, the narrative layer itself involved interpretive ambiguity. A span could be narrative background, an opening hook, claim presentation, or supporting evidence depending on how its rhetorical role was interpreted by the one annotating. This is why the narrative Gamma results should be interpreted more cautiously than the argumentation results.

The model-model results also reveal an important pattern. LLMs often agreed more with each other than with humans, especially in the narrative layer. This suggests that the models shared annotation tendencies that differed from the human annotation strategy. These tendencies may include selecting larger coherent discourse sections, relying on the surface position of spans, or treating narrative flow as more important than the functional distinctions defined in the annotation guidelines.

The results therefore support the broader claim of the thesis: narrativity affects annotation and model performance when increasing the interpretive complexity of span selection and rhetorical-function classification. In explicitly argumentative TED spans, human annotators could identify claims and premises consistently, but models still struggled to match human span and label decisions. In narrative-function annotation, even human agreement was low, showing that the interpretation of narrative role within persuasive and argumentative discourse is more ambiguous.

Overall, the Gamma analysis shows that the argumentation layer provides a stronger basis for model evaluation than the narrative layer. The high human-human agreement in argumentation means that model-human disagreement can be interpreted as a meaningful indication of model difference from human annotation. The narrative layer, by contrast, should be treated as exploratory: it remains useful for understanding how models handle narrative structure, but its low human-human agreement means that narrative-layer results should not be interpreted as a stable gold-standard evaluation in the same way.

5.2 LLM PROMPTING AND EVALUATION RESULTS

This section presents the main LLM evaluation results for the TED and STORAL datasets. The results are discussed together with observations from the prompt-development process because the final model outputs were strongly shaped by the type of prompt used for each task. The TED corpus was evaluated through span-based F1 metrics, while the STORAL corpus was evaluated through semantic similarity metrics and supporting-span overlap. This difference reflects the different nature of the two tasks: TED annotation required the extraction of explicit argumentative or narrative spans, whereas STORAL annotation required the reconstruction of implicit argumentative meaning through explicit formulations.

The final prompts differed across the three annotation tasks. For TED argumentation, the final prompt used a few-shot structure with detailed definitions of MC-EXP, C-EXP, and P-EXP, together with positive and negative examples. For TED narration, the final prompt used a zero-shot structure. It contained detailed definitions of the rhetorical-function labels, such as Exordium, Narratio, Propositio, Confirmatio, Peroratio, and Other, but did not include examples. For STORAL, the final prompt used a few-shot structure because the task required the model not only to identify a label, but also to select a supporting span and generate an explicit formulation of an implicit argument component.

A general observation from prompt development was that the most useful prompts were those that reproduced the human annotation guidelines as closely as possible. In particular, the final prompts did not simply ask the models to "find arguments" or "find narratives". Instead, they gave the models detailed label definitions, decision rules, span-selection rules, and output-format constraints. This is consistent with work on prompt engineering and in-context learning, which shows that LLM performance can be highly sensitive to the wording, structure, and specificity of the prompt (Li, 2023; Chen et al., 2025). It also supports the idea that prompt optimization is model- and task-dependent rather than universal (Chen et al., 2023).

Table 5.5: Comparison between the last tested prompt and the final evaluation results for the TED corpus. Scores are averaged across the two human annotators and across relaxed F1, overlap F1, character-offset F1, and sentence-level F1.

Layer	Model	Last	Final	Diff.	Prompt design
Argumentation	Gemma	0.499	0.330	-0.169	Few-shot prompt with detailed label definitions and examples.
Argumentation	Qwen	0.363	0.244	-0.119	Few-shot prompt with detailed label definitions and examples.
Argumentation	Mistral	0.323	0.243	-0.080	Few-shot prompt with detailed label definitions and examples.
Narration	Gemma	0.460	0.438	-0.022	Zero-shot prompt with detailed rhetorical-function definitions, but no examples.
Narration	Qwen	0.366	0.445	+0.078	Zero-shot prompt with detailed rhetorical-function definitions, but no examples.
Narration	Mistral	0.415	0.364	-0.051	Zero-shot prompt with detailed rhetorical-function definitions, but no examples.

Table 5.5 compares the last tested prompt results with the final evaluation results for the TED corpus. These values should not be interpreted as a direct before-and-after improvement score, because the last tested prompt was evaluated on the prompt-

development subset, while the final results were calculated on the final evaluation subset. However, the comparison is still useful because it shows how performance changed when the same optimized prompt was applied to unfamiliar final texts.

For TED argumentation, all three models performed lower on the final evaluation set than on the prompt-development set. Gemma decreased from an average F1 of 0.499 to 0.330, Qwen decreased from 0.363 to 0.244, and Mistral decreased from 0.323 to 0.243. This suggests that the argumentation prompt was able to produce stronger results on the texts used during prompt development, but that performance did not fully generalize to the final evaluation set. The drop was largest for Gemma, which had the strongest result in the tested stage but also the largest decrease in the final evaluation.

The narration layer showed a different pattern. Gemma and Mistral decreased slightly from the tested subset to the final subset, while Qwen improved from 0.366 to 0.445. This suggests that the zero-shot narrative prompt was more stable across the two sets than the argumentation prompt. A notable finding from prompt development was that the best final narration prompt did not use examples. Instead, it relied on detailed rhetorical-function definitions and decision rules. This contrasts with the argumentation layer, where examples were useful for showing the model how to distinguish argumentative components from non-argumentative material.

This difference between the two layers suggests that examples do not help all annotation tasks in the same way. In the argumentation task, few-shot examples helped illustrate the difference between claims, premises, and non-argumentative text. In the narrative task, however, examples could have encouraged the model to imitate particular narrative structures too closely. Since narrative-function annotation depends strongly on the rhetorical role of a span within the whole talk, detailed definitions and decision rules appear to have been more useful than examples.

Table 5.6: Final TED evaluation results by layer and model. Scores are averaged across the two human annotators.

Layer	Model	Relaxed F1	Overlap F1	Char.-offset F1	Sentence F1	Average
Argumentation	Gemma	0.322	0.331	0.319	0.350	0.330
Argumentation	Qwen	0.245	0.231	0.252	0.248	0.244
Argumentation	Mistral	0.241	0.241	0.237	0.252	0.243
Narration	Qwen	0.469	0.417	0.436	0.457	0.445
Narration	Gemma	0.447	0.417	0.452	0.435	0.438
Narration	Mistral	0.362	0.320	0.376	0.399	0.364

Table 5.6 shows the final TED evaluation results. In the argumentation layer, Gemma achieved the strongest average performance, with an average F1 of 0.330. Qwen and Mistral performed similarly, with averages of 0.244 and 0.243. Although Gemma was the strongest argumentation model, the scores remain relatively low overall. This indicates that the models did not consistently reproduce the human span and label decisions, even though the task focused only on explicit argumentative components.

The argumentation scores are particularly important when interpreted together with the Gamma results. The human-human Gamma agreement for the argumentation layer was high, which suggests that the human annotations formed a relatively stable reference. Therefore, the low model-human F1 scores cannot be explained simply by instability in the human annotations that were used as reference. Rather, they indicate that the models often selected different spans, missed human-annotated components, or assigned different labels from the human annotators.

The narration layer produced higher final F1 scores than the argumentation layer. Qwen achieved the highest narration score, with an average F1 of 0.445, followed

closely by Gemma at 0.438. Mistral was lower, with an average of 0.364. These results suggest that the models were more successful at identifying broad narrative material than at extracting explicit argumentative components. However, this does not necessarily mean that the models fully solved the narrative-function task. The qualitative error analysis, which will dive deeper into more detail in Section 5.3, showed that the models often selected large coherent narrative blocks and sometimes confused rhetorical functions. Therefore, the higher narration F1 scores should be interpreted alongside the low human-human Gamma agreement for the narrative layer, which shows that this layer was itself more interpretive and less stable.

Across TED results, one pattern is clear: the models performed better when the task involved recognizing broader discourse material than when it required precise argumentative component extraction. Argumentation annotation required the models to distinguish major claims, claims, and premises, while also selecting minimal spans. Narration annotation allowed larger coherent discourse units, which may have been easier for the models to identify. However, the models still struggled with functional interpretation, especially when the same passage could introduce a topic, provide background, or serve as evidence.

Table 5.7: Comparison between the last tested prompt and the final evaluation results for the STORAL corpus. SBERT measures semantic similarity between gold and model-generated explicit formulations. BERTScore F1 is calculated using the same reference-candidate formulation comparison.

Model	SBERT last	SBERT final	Diff.	BERTScore last	BERTScore final	Diff.
Qwen	0.635	0.485	-0.150	0.436	0.326	-0.110
Gemma	0.577	0.470	-0.107	0.421	0.333	-0.088
Mistral	0.475	0.461	-0.014	0.350	0.341	-0.009

Table 5.7 shows the difference between the last tested STORAL prompt results and the final STORAL evaluation results. As with the TED results, these scores come from different subsets, therefore, the decrease in scores should not be read as a direct decline in prompt quality. Instead, they show how the prompt behaved when applied to the final evaluation stories.

For STORAL, all three models achieved lower semantic similarity scores on the final evaluation set than on the prompt-development set. Qwen had the highest SBERT score in the last tested stage, 0.635, but decreased to 0.485 in the final evaluation. Gemma decreased from 0.577 to 0.470, while Mistral remained relatively stable, decreasing only from 0.475 to 0.461. The BERTScore F1 results show the same pattern. Qwen and Gemma decreased more noticeably, while Mistral changed only slightly.

This result suggests that the prompt-development stories may have been easier for Qwen and Gemma than the final evaluation stories, or that these models were more sensitive to variation between stories. Mistral had lower peak performance in the tested stage, but its final performance was more stable. This is important because it shows that the best-performing model during prompt testing is not necessarily the model that generalizes the best to unfamiliar final texts.

Table 5.8: Final STORAL evaluation results by model. The supporting-span F1 values differ slightly across SBERT-based and BERTScore-based matching because the best human-model annotation pairings are selected separately for each metric.

Model	SBERT	BERTScore F1	Span F1, SBERT match	Span F1, BERTScore match	Matches
Qwen	0.485	0.326	0.659	0.584	22
Gemma	0.470	0.333	0.569	0.564	32
Mistral	0.461	0.341	0.450	0.464	24

Table 5.8 presents the final STORAL evaluation results by model. The differences between the models are relatively small for semantic similarity. Qwen obtained the highest SBERT score, 0.485, followed by Gemma at 0.470 and Mistral at 0.461. In contrast, Mistral obtained the highest BERTScore F1, 0.341, followed by Gemma at 0.333 and Qwen at 0.326. This difference between SBERT and BERTScore suggests that the ranking of the models depends on the semantic metric used.

The supporting-span scores show a different pattern. Qwen had the highest supporting-span F1 when pairings were based on SBERT similarity, with a score of 0.659. Gemma followed with 0.569, and Mistral had the lowest score at 0.450. Under BERTScore-based matching, Qwen again had the highest supporting-span F1, 0.584, followed by Gemma at 0.564 and Mistral at 0.464. This suggests that Qwen was particularly strong at selecting textual evidence that overlapped with the gold supporting spans, even when its generated formulations were not always the closest according to BERTScore.

The STORAL scores are not directly comparable to the TED F1 scores because they measure different aspects of model performance. TED evaluation focused on extracted spans and labels, while STORAL evaluation focused on generated explicit formulations and supporting-span overlap. However, the STORAL results show a similar broader issue: the models were able to produce plausible annotations, but they did not consistently match the human reference interpretation.

Table 5.9: Final STORAL semantic evaluation by model and implicit argument label.

Model	Label	SBERT	BERTScore F1	Matches
Gemma	C-IMP	0.164	0.192	1
Gemma	MC-IMP	0.540	0.381	14
Gemma	P-IMP	0.430	0.302	17
Mistral	MC-IMP	0.465	0.365	14
Mistral	P-IMP	0.454	0.307	10
Qwen	C-IMP	0.272	0.107	1
Qwen	MC-IMP	0.537	0.363	14
Qwen	P-IMP	0.411	0.282	7

Table 5.9 shows that implicit major claims were generally easier for the models than implicit premises. For Gemma and Qwen, MC-IMP received the highest SBERT scores, 0.540 and 0.537 respectively. Mistral showed a smaller difference between MC-IMP and P-IMP, with scores of 0.465 and 0.454. The BERTScore results show a similar pattern: MC-IMP generally received higher scores than P-IMP.

This pattern is consistent with the qualitative error analysis, which is explained later in more detail. The models often recovered the global moral or takeaway of a story more successfully than the intermediate reasoning claims. In STORAL stories, the implicit major claim usually corresponds to the most recognizable lesson of the narrative. By contrast, implicit premises require the model to reconstruct the reasoning that connects specific events to the moral. This is a more demanding task because the model must identify not only what the story means overall, but how particular events support that meaning.

The C-IMP results should be interpreted cautiously because there was only one matched C-IMP annotation for Gemma and one for Qwen. These scores therefore cannot support a general conclusion about model performance on implicit claims. Instead, the main label-level contrast should be drawn between MC-IMP and P-IMP, where more matches were available.

Overall, the LLM evaluation results show that model performance depended strongly on task type, prompt structure, and discourse type. In TED argumentation, the

models struggled to reproduce human span and label choices even though the argumentation layer had a stable human-human baseline. In TED narration, the models achieved higher F1 scores, but those scores must be interpreted cautiously because narrative-function annotation was less stable between human annotators. In STORAL, the models often generated plausible moral interpretations, but semantic similarity scores show that these interpretations did not always match the gold annotation formulations.

The prompt-development results also show that different annotation tasks benefited from different prompt designs. For TED argumentation, few-shot examples were useful because they helped define what should and should not count as an argumentative component. For TED narration, the best final prompt didn't use examples and instead relied on detailed functional definitions and decision rules. For STORAL, few-shot examples were important because the task required the model to perform multiple operations at once: identify an implicit component, select a supporting span, and generate an explicit formulation. These differences support the methodological decision to treat prompt optimization as task-specific rather than applying one uniform prompt structure to all layers and datasets.

5.3 QUALITATIVE ERROR ANALYSIS OF TED TALKS

A qualitative error analysis was conducted on four TED talks selected from the final evaluation set. The purpose of this analysis was to examine the types of errors that occurred behind the quantitative scores and to identify recurring patterns in the LLM-generated annotations. The four selected talks were TED_002, TED_008, TED_010, and TED_011. These texts were chosen to represent different performance profiles across the argumentation and narration layers. TED_002 was selected as the strongest overall case, since it received relatively high F1 scores for both argumentation and narration. TED_010 and TED_008 were selected as mixed cases: TED_010 showed weak argumentation performance but stronger narration performance, while TED_008 showed the opposite pattern, with stronger argumentation performance but weaker narration performance. TED_011 was selected as a low-performing case, especially for the narrative layer.

Table 5.10: Summary of the four TED talks selected for qualitative error analysis.

Text	Argumentation F1	Narration F1	Main qualitative pattern
TED_002	0.448	0.646	Highest overall case. Models identified explicit claims and historical evidence relatively well, but still showed boundary expansion and major-claim/claim confusion.
TED_008	0.388	0.247	Mixed case. Argumentation was easier because the talk contained repeated explicit claims, but narrative-function annotation was difficult because stories also functioned as evidence.
TED_010	0.080	0.505	Mixed case. The models detected narrative anecdotes more successfully than argumentative components, because the argument was embedded in comedy and indirect persuasion.
TED_011	0.302	0.108	Low narrative case. The models found some explicit claims but struggled with nested stories, explanatory examples, and rhetorical-function segmentation.

The error analysis used the detailed span-comparison files generated during evaluation. These files contained, for each model-human comparison, the human label, the LLM label, the human span, the LLM span, the overlap score, the intersection-over-union value, and the automatic match status. Since both human annotators were included as references, the analysis did not treat one annotator as the only possible gold standard. Instead, disagreement with both annotators was interpreted as stronger evidence of a model error, while disagreement with only one annotator was treated more cautiously, since it may also reflect annotation ambiguity.

The manual inspection focused on four broad error categories: missed human annotations, extra model annotations, span-boundary errors, and label-confusion errors. For the argumentation layer, label-confusion errors mainly involved confusion between major claims, claims, and premises. For the narrative layer, label-confusion errors involved confusion between the rhetorical narrative functions *Exordium*, *Narratio*, *Propositio*, *Confirmatio*, and *Peroratio*. Near-exact matches were not counted as errors, although some same-label overlaps were still inspected when the LLM span was substantially broader or narrower than the human span.

Table 5.11: Manual error categories across the three model-human comparison instances across three models and two human annotators. Near-exact matches are excluded from the error counts.

Error category	Arg. n	Arg. %	Narr. n	Narr. %	Interpretation
Missed human annotation	328	58.3	161	57.5	A reference annotation produced by a human annotator was not recovered by the model.
Extra model annotation	136	24.2	34	12.1	The model produced an annotation for which there was no corresponding human annotation.
Span-boundary/segmentation error	35	6.2	59	21.1	The model identified related material but selected a span that was too broad, too narrow, or differently segmented.
Label/function confusion	64	11.4	26	9.3	The model selected overlapping material but assigned a different label from the human annotator. In argumentation, this involved claim/premise distinctions; in narration, this involved rhetorical-function labels.
Total coded errors	563	100.0	280	100.0	

Table 5.11 shows that the most frequent error type in both layers was the missed human annotation. In the argumentation layer, missed annotations accounted for 58.3% of coded errors, while in the narrative layer they accounted for 57.5%. This suggests that the models more often failed to identify relevant human-annotated material than they selected completely wrong spans. This pattern is especially important because it indicates that the models were not simply over-generating annotations, rather, they frequently failed to capture the full density of human annotation, particularly when argumentative or narrative functions were distributed across several small spans.

The second most common error type differed by layer. In the argumentation layer, extra model annotations were the second-largest category, accounting for 24.2% of

coded errors. This indicates that the models sometimes interpreted persuasive or explanatory statements as argumentative components even when they were not annotated as such by the human annotators. In the narrative layer, however, span-boundary and segmentation errors were the second-largest category, accounting for 21.1% of coded errors. This reflects a different difficulty: for narrative-function annotation, the models often identified relevant stretches of discourse but selected spans that were too large or that combined several rhetorical functions into one annotation.

TED_002: High-performing case

TED_002 was the strongest overall case in the selected sample. Its average F1 was 0.448 for argumentation and 0.646 for narration. The talk explains the history and social consequences of IQ testing. The structure of the talk is relatively explicit: it introduces the historical origin of IQ tests, describes how they were misused in eugenic and discriminatory contexts, and ends with a clear evaluative conclusion that IQ tests should not be treated as a measure of a person's overall potential. This relatively explicit organization helps explain why the models performed better on this text than on the other selected cases.

In the argumentation layer, the models often identified central argumentative statements correctly. For example, the statement "IQ test results have been used to justify horrific policies and scientifically baseless ideologies" was matched closely by the models and corresponds to a central important claim in the talk. Similarly, the statement "What started as a way to identify those who needed academic help quickly became used to sort people in other ways" was also annotated as a claim component. These examples show that the models were relatively successful when the argumentative function was explicitly stated and expressed in a simple declarative sentence.

However, TED_002 also shows that strong overall performance does not mean that the models' annotations were identical to the human annotations. One recurring error was label confusion between major claims and ordinary claims. For example, the concluding statement that researchers "reject the notion that individuals can be categorized by a single numerical score" was labelled by a model as a claim rather than a major claim. This is a meaningful disagreement because the sentence does not simply support another claim, but it functions as the final overall position of the talk. The error therefore reflects difficulty in distinguishing local argumentative statements from the global thesis of the text.

Another recurring pattern in TED_002 was boundary expansion. In several cases, the model selected the correct argumentative material but included surrounding context. For instance, the human annotation selected the premise about forced sterilization in Virginia, while the model included the continuation that the decision was upheld by the Supreme Court. This is not a severe semantic error, since the extra material remains relevant evidence, but it affects span-level evaluation because the model does not take into account the same minimal annotation boundary.

The narrative layer in TED_002 was also relatively strong, but the errors were mostly related to segmentation. The models often identified historically explanatory sections as *Confirmatio*, which is understandable because the historical evidence supports the talk's argument. However, they sometimes selected large blocks that combined several rhetorical functions. For example, sections introducing the origin of IQ testing and later sections describing the misuse of IQ testing were sometimes grouped into long narrative spans. This indicates that the model understood the general historical development but did not always separate background narration from argumentative evidence in the same way as the human annotators.

Overall, TED_002 suggests that LLMs can perform relatively well when argumentative claims are explicit and when the discourse follows a clear chronological-explanatory structure. The remaining errors are mostly fine-grained: distinguishing major claim from claim, selecting minimal spans, and separating background information from supporting evidence.

TED_008: Mixed case with stronger argumentation than narration

TED_008 was selected as a mixed case because its argumentation score was relatively stronger than its narrative score. The talk argues that people must define and communicate their value in order to be paid appropriately. It contains a highly explicit central thesis, repeated near the beginning and again at the end: "No one will ever pay you what you're worth. They'll only ever pay you what they think you're worth". This explicit repetition made the argumentative structure easier for the models to identify.

In the argumentation layer, the models often recovered important claims and premises. The central idea that communicating one's value is essential to being paid well was generally identified, and the models also captured several supporting examples, such as the evidence that a woman employee earns less than a man and that women business owners underprice their work. The models were also able to identify outcome-based premises, such as the statement that the woman in the second story is now charging three times as much for web design.

The main argumentation errors in TED_008 were span-boundary errors and missed smaller argumentative components. In several cases, the model selected a broader span than the human annotator. For example, the human annotation selected "She's now charging three times as much for web design" as a premise, while the model selected a longer span including both the increase in price and the fact that the business was growing. This is semantically related and can be argued that it can be considered an evidence, but it differs from the more minimal human span. Another example is the claim "they shouldn't limit our earning potential", where the model included the preceding sentence about doubts and fears being natural. These boundary choices show a tendency to include explanatory context around concise argumentative statements.

The narrative layer was more difficult for the models. TED_008 contains two personal/business stories that function as evidence for the broader argument. The models often recognized these stories as relevant, but they did not consistently assign the same rhetorical function as the human annotators. A particularly clear example is the opening thesis of the talk. One human annotation treated the opening statement as *Propositio* because it presents the central claim of the talk, while one model labelled a largely overlapping span as *Exordium*. This disagreement shows that the model sometimes relied on position in the talk rather than rhetorical function. Because the span occurs at the beginning, the model treated it as an introduction, even though functionally, it represents the central argumentative proposition.

A similar issue occurred with the two business-owner stories. These stories were sometimes treated by the models as *Confirmatio* because they provide evidence for the speaker's argument. However, some human annotations treated parts of these story introductions as *Propositio*, especially where the speaker explicitly announces that she will tell a story about communicating value. This type of disagreement reflects a difficulty in the narrative annotation scheme: a story can introduce a case, provide evidence, and advance the central argument at the same time. The models often captured the broad relevance of the story but struggled to assign a single precise rhetorical function.

TED_008 therefore shows that explicit argumentative claims can be easier for the models than narrative-function distinctions. The models were often able to identify what the speaker was arguing, but they were less consistent in distinguishing whether a story segment functioned as background, claim presentation, or supporting evidence.

TED_010: Mixed case with weaker argumentation but stronger narration

TED_010 was selected as the opposite mixed case. Its argumentation score was very low, with an average F1 of 0.080, while its narrative score was substantially higher at 0.505. The talk is a stand-up comedy performance about stereotypes, Middle Eastern identity, travel, and representation. This makes the argumentation layer difficult because much of the persuasive meaning is expressed indirectly through jokes, anecdotes, and humorous contrasts rather than through explicit argumentative statements.

The models identified a few central argumentative ideas. For example, the statement "there's good people everywhere" was recognized as argumentatively important, and the statement "I try, with my stand-up, to break stereotypes, present Middle Easterners and Muslims in a positive light" was also recovered by some models. However, these were among the few relatively explicit claims in the talk. Majority of the remaining argumentative context depends on comedic examples. The speaker uses personal experiences at borders, Hollywood casting stereotypes, terrorism-related news coverage, and travel anecdotes to challenge stereotypes about Middle Easterners and Muslims. These passages are persuasive, but they do not always take the form of explicit claim-premise structures.

A recurring argumentation error in TED_010 was label confusion between major claims, claims, and premises. For instance, "there's good people everywhere" was labelled by a human annotator as a major claim, but models sometimes labelled it as a claim. Similarly, "All it takes in one person to mess it up" was sometimes treated as a premise rather than a claim. These errors are understandable because the talk does not present its thesis in a formal argumentative structure. Instead, claims emerge through punchlines and narrative framing. The models therefore tended to identify locally important statements but did not always assign them the correct argumentative label.

There is a high number of missed argumentation annotations in TED_010 which is especially revealing. Human annotations included statements such as "being Iranian-American presents its own set of problems" and "some of the Arab countries aren't getting along with Iran either". These statements are important because they establish the social and political tensions that motivate the speaker's argument. However, because they are embedded in humorous storytelling, the models often failed to treat them as argumentative components. This supports the broader finding that argument extraction becomes more difficult when persuasive meaning is indirect, humorous, or strongly narrative.

The narrative layer was stronger for TED_010 because the talk is organized around clear anecdotes. The models could identify large story-based units, such as the Kuwait border-control anecdote, the Hollywood casting anecdote, and the Dubai travel anecdotes. However, the main narrative errors involved overly broad spans. For example, a human annotator selected a specific Kuwait border episode as Narration, while the model selected a much longer opening section that included the founding of the comedy tour, the speaker's Iranian-American identity, passport issues, and the Kuwait story. This suggests that the model recognized the narrative nature of the opening but did not split it into the smaller functional units used by the human annotators.

There were also narrative-function confusions. For example, the opening description of the Axis of Evil Comedy Tour was labelled as Exordium by a human annotator but as Narratio by the model. This is a typical rhetorical-function error: the same span is both introductory and narrative. The model seemed to have prioritized the fact that the speaker was telling background information about himself, whereas the human annotation treated the passage as an opening to the talk. Similarly, some Kuwait border-control talk was labelled as Narratio by a human annotator but as Confirmatio by the model, because the anecdote also functions as evidence for the speaker's point about stereotyping and identity.

TED_010 therefore is a great example for this thesis especially. The models were better at detecting narrative material than extracting argument components from that narrative material. In this text, the persuasive structure is not absent, but it is embedded in comedy and anecdotal storytelling. As a result, the LLMs often recognized that a story was being told but did not consistently recover the argumentative function of that story.

TED_011: Low-performing case, especially for narration

TED_011 was selected as a low-performing case, particularly for the narrative layer. The talk argues that science communication should use storytelling, accessible language, and emotional engagement rather than excessive "jargon" and seriousness. The text begins with a classroom anecdote, then presents a story-based explanation of viruses, and later uses this example to argue against what the speaker calls the "tyranny of precision" in science education.

In the argumentation layer, the models recovered some explicit claims accurately. For example, statements such as "good storytelling is all about emotional connection" and "science has become that horrible storyteller that we all know who gives us all the details nobody cares about" were matched closely. These are concise, explicit, evaluative statements, and therefore resemble the kinds of claims that LLMs handled better in the other TED talks as well.

However, TED_011 also produced many missed annotations. The models often failed to annotate premises embedded in the classroom story and the science-explanation example. Human annotations included statements such as "I didn't understand a word of it", "the language in their textbook was truly incomprehensible", and "it would be accurate, but it would be completely impossible to understand". These statements support the speaker's argument about inaccessible science communication, but they are embedded in narrative and explanatory passages. The models therefore often treated them as part of the story or explanation rather than as argumentative evidence.

There were also claim-premise confusions. For example, the statement "if a young learner can't understand anything in science and learns to hate it because it all sounds like this, that will ruin their chances of success" was labelled as a premise by a human annotator but as a claim by a model. Similarly, "leave out the seriousness, leave out the jargon" was labelled as a major claim by a human annotator but as a claim by the model. These examples show that the models recognized the argumentative relevance of the spans, but they did not consistently distinguish between supporting reasons, local claims, and the main thesis.

The narrative layer was particularly difficult in TED_011 because the talk contains multiple nested narrative and explanatory structures. The opening classroom anecdote functions as an introduction to the problem, the virus story functions as a teaching example, and the later discussion of science writing functions as argumentative explanation. The models often selected large spans that combined these functions. For example, the opening Exordium was expanded by some models to include not only the initial teaching situation but also the textbook-reading episode

and the student's reaction. This produced low overlap with the human annotation even though the selected material was contextually related.

Narrative-function confusion was also common. The virus story was sometimes labelled as *Confirmatio* by the models, whereas parts of it were labelled as *Exordium* by a human annotator. This reveals an important ambiguity: the virus story is narratively presented, but it also functions as evidence for the speaker's claim that storytelling improves science communication. The models tended to interpret such passages according to their evidential role, while the human annotations sometimes emphasized their function in structuring the speaker's explanation.

TED_011 therefore shows that the narrative layer can be especially difficult when the text contains stories within explanations. The models did not simply fail because the text lacked structure. Rather, the difficulty came from the fact that one passage could simultaneously introduce a topic, illustrate a scientific concept, and support a broader argument about communication. This produced both boundary errors and rhetorical-function confusions.

Table 5.12: Representative examples from the TED error analysis.

Text	Layer	Pattern	Example and interpretation
TED_002	Argumentation	Major claim vs. claim	The final statement rejecting the idea that people can be categorized by a single numerical score was sometimes labelled as a claim rather than a major claim. The model identified the relevant passage but underestimated its global argumentative role.
TED_002	Narration	Boundary expansion	Historical evidence about eugenics and IQ testing was often selected as one large Confirmatio span. The model captured the evidential function but did not segment the historical evidence as narrowly as the human annotators.
TED_008	Narration	Propositio vs. Exordium	The opening statement "No one will ever pay you what you're worth" was treated by one model as Exordium, although it functions as the central proposition of the talk. This suggests reliance on textual position rather than rhetorical function.
TED_008	Argumentation	Broad premise span	The premise "She's now charging three times as much for web design" was often expanded to include additional outcome information. This produced a related but broader evidence span.
TED_010	Argumentation	Indirect argument missed	Statements about stereotyping and Middle Eastern identity were frequently missed because they were embedded in jokes and anecdotes rather than presented as formal claims or premises.
TED_010	Narration	Narratio vs. Confirmatio	The Kuwait border-control anecdote was sometimes labelled Narratio and sometimes Confirmatio. The model recognized the story, but the story also functioned as evidence for the argument about stereotyping.
TED_011	Argumentation	Claim vs. Premise	The statement that inaccessible science language can ruin a learner's chances of success was treated as a claim by a model but as a premise by a human annotator. The model recognized argumentative relevance but not the support relation.
TED_011	Narration	Nested-story segmentation	The opening teaching anecdote and the virus explanation were often merged into longer spans. This shows difficulty separating introductory narrative, illustrative story, and argumentative evidence.

Cross-text error patterns

Across the four TED talks, the most consistent error pattern was missed human annotations. This occurred in both layers and across all selected texts. In the argumentation layer, missed annotations often involved smaller claims and premises embedded inside stories, examples, or explanatory passages. The models were more successful when claims were explicitly stated as general propositions, such as "good storytelling is all about emotional connection" or "IQ test results have been used to justify horrific policies". When argumentative meaning relied on the function of a narrative piece within broader discourse, they performed worse.

The second major pattern was over-selection of span boundaries done by the models. This was especially common in the narrative layer. The models often selected long, coherent discourse blocks rather than the shorter rhetorical-function spans selected by human annotators. This suggests that the models were good at recognizing when a section of text stayed on the same topic or followed a continuous story, but they struggled to break it down into smaller parts based on specific rhetorical roles, as required by the annotation guidelines.

A third pattern was label confusion. In the argumentation layer, the most common label confusions involved claims and premises, or major claims and claims. These errors show that the models could often detect argumentative relevance but had difficulty assigning hierarchical argumentative status. In the narrative layer, label confusion often involved Exordium, Narratio, Propositio, and Confirmatio. These categories are functionally defined rather than positionally defined, and the models sometimes appeared to rely on surface position or broad discourse type instead of rhetorical function.

Finally, some disagreements should be interpreted as meaningful non-errors or borderline cases. This is especially important for narrative-function annotation. A personal story can function as Narratio because it tells a story, but it can also function as Confirmatio when it provides evidence for the speaker's claim. Similarly, an opening passage can function as Exordium because it begins the talk, but it can also function as Propositio if it presents the central claim. These cases show that low agreement does not always mean that the model output is random or irrelevant. In some cases, the model selected an acceptable interpretation, but not the same interpretation used by the human annotator.

Overall, the TED error analysis shows that LLMs are better at identifying explicit, self-contained argumentative statements than at recovering argumentative structure embedded in narrative discourse. The narrative layer presents a different challenge: models can often recognize narrative material, but they struggle to segment it according to rhetorical function. These findings support the broader argument of the thesis that narrativity affects LLM performance not only by making arguments more implicit, but also by increasing the number of possible functional interpretations of the same textual span.

5.4 QUALITATIVE ERROR ANALYSIS OF STORAL STORIES

A second qualitative error analysis was also conducted on four STORAL stories selected from the final evaluation set. Unlike the TED evaluation, where the model outputs were compared against two separate human annotators, the STORAL evaluation used one shared gold annotation created through agreement between the two annotators. This made the STORAL error analysis more direct: each model output was compared with the same gold reference annotation for each story.

These stories were selected to represent different performance profiles across the semantic similarity and supporting-span metrics. STORAL_011 was selected as a high-performing case, because it had the strongest overall semantic alignment and relatively strong supporting-span overlap. STORAL_008 and STORAL_007 were selected as mixed cases. STORAL_008 had relatively high supporting-span overlap but low semantic similarity, making it useful for examining cases where the model selected relevant evidence but formulated the implicit meaning differently from the gold annotation. STORAL_007 showed relatively high SBERT and supporting-span scores but lower BERTScore F1, making it useful for examining differences between semantic evaluation metrics. Lastly, STORAL_005 was selected as a low-performing case, because it had low SBERT similarity, low BERTScore F1, and low supporting-span F1.

Table 5.13: Summary of the four STORAL stories selected for qualitative error analysis.

Text	SBERT	BERTScore F1	Span F1	Main qualitative pattern
STORAL_011	0.620	0.488	0.671	High-performing case. Models recovered the global moral about facing problems, but often missed the implicit premises.
STORAL_008	0.270	0.284	0.675	Mixed case. Models often found relevant evidence but formulated the moral as cleverness overcoming strength rather than flattery and pride causing trouble.
STORAL_007	0.607	0.292	0.655	Mixed case. Models produced semantically related loyalty-based formulations, but BERTScore was lower because the wording and abstraction level differed from the gold annotation.
STORAL_005	0.293	0.243	0.333	Low-performing case. Models inferred different lessons about deception, conflict, or carelessness instead of the gold meaning about blind trust.

The STORAL error analysis focused on implicit argumentation. Each gold annotation contained a label, a supporting span, and an explicit formulation of the implicit argumentative meaning. The LLM outputs were evaluated in the same format. Therefore, the analysis did not only consider whether the model selected the correct textual evidence, but also whether the model-generated formulation expressed the same implicit meaning as the gold annotation. This is important because a model may identify the correct story span while drawing a different moral or inference from it.

The manual inspection focused on seven categories: missed implicit components, semantically weak formulations, partial semantic alignment, correct supporting span with weak formulation, semantically related formulation with weak supporting span, extra model annotations, and meaningful near matches. The category "meaningful near match" was included because some model outputs were close to the gold annotation even when the metric scores were not perfect. These cases were not treated as clear errors, but they were kept in the table in order to show that not all disagreements should be interpreted as model failure.

Table 5.14: Manual error categories in the STORAL qualitative error-analysis sample. Counts refer to model-gold comparison instances across three models.

Category	Count	%	Interpretation
Missed implicit component	17	39.5	A gold implicit component had no corresponding LLM formulation for the same label.
Semantically weak formulation	9	20.9	The model produced an implicit formulation, but the formulation expressed a different moral or inference from the gold annotation.
Partial semantic alignment	5	11.6	The model captured part of the intended implicit meaning but missed an important element of the gold formulation.
Correct supporting span, weak formulation	3	7.0	The model selected highly overlapping textual evidence but formulated a different implicit meaning.
Semantically related, weak supporting span	3	7.0	The model formulation was related to the gold interpretation, but its supporting span did not align closely with the gold span.
Extra model annotation	1	2.3	The model produced an additional implicit component that was not present in the gold annotation.
Meaningful near match	5	11.6	The model output was close enough to the gold annotation to be treated as a meaningful alignment or borderline non-error.
Total	43	100.0	

Table 5.14 shows that the most frequent category in the STORAL error-analysis sample was the missed implicit component. This accounted for 39.5% of the inspected comparison annotations. This pattern is different from a purely surface-level extraction problem: the models did not only struggle to copy the correct supporting span, but often failed to produce one of the implicit argumentative components at all. This was especially visible for implicit premises, which were often more specific than the global moral of the story.

The second largest category was semantically weak formulation, which accounted for 20.9% of the inspected cases. These were cases where the model produced an implicit statement, but the statement did not correspond closely to the gold formulation. In many cases, the model inferred a plausible moral from the story but not the same moral that was represented in the gold annotation. This is especially important for STORAL because the task involved reconstructing implicit argumentation. A formulation can be grammatically well-formed and even story-related while still failing to express the intended argumentative meaning.

The category of meaningful near matches accounted for 11.6% of the inspected cases. These examples show that some disagreements were not clear model errors. In several cases, the model produced a formulation that was semantically close to the gold annotation but used different wording or a different level of generality. This category is particularly relevant for interpreting the difference between SBERT and BERTScore. SBERT sometimes assigned relatively high similarity to formulations that expressed a similar general idea, while BERTScore was lower when the wording was substantially different.

STORAL_011: High-performing case

STORAL_011 was selected as the high-performing case. The story describes Swami Vivekananda being chased by monkeys on a narrow path. At first, he tries to run away, but an old sanyasi tells him to face them. When he turns around and faces the

monkeys, they retreat. The shared gold annotation formulated the implicit major claim as: "you can overcome your problems by facing them with courage".

This story produced the strongest overall alignment in the STORAL sample. The average SBERT score was 0.620, the average BERTScore F1 was 0.488, and the average supporting-span F1 was 0.671. All three models recovered the implicit major claim relatively well. Gemma formulated it as "Facing your fears with courage is the most effective way to overcome them", Mistral formulated it as "Facing your fears can make them disappear", and Qwen formulated it as "Confronting challenges directly is more effective than running away from them". These formulations differ in wording, but all preserve the main implicit meaning of the gold annotation: problems or fears should be faced directly rather than avoided.

However, even in this high-performing story, the models missed several implicit premises. One gold premise was "you are trying to escape your problems", supported by the span in which Swami Vivekananda begins to run and the monkeys chase him. This premise was missed by all three models. Another gold premise was "you realize that trying to run away doesn't solve them", supported by the moment when the old sanyasi tells him to face the monkeys and he stops running. Gemma produced a similar formulation, but it shifted the interpretation toward the attacker: "Running away from a threat often increases the aggression of the attacker". This is connected to the story, but it does not fully match the gold premise, which focuses on the protagonist's realization that trying to escape is useless in this situation.

This case shows that LLMs can successfully infer a global implicit moral when the narrative has a clear resolution and a recognizable lesson. However, the models were less consistent in identifying the smaller implicit premises that support that moral. In other words, the models often captured the final takeaway of the story but did not always reconstruct the intermediate reasoning steps.

STORAL_008: Mixed case with high span overlap but weak formulation

STORAL_008 was selected as a mixed case because the models often selected relevant supporting spans but formulated a different implicit meaning from the gold annotation. The story describes a lamb captured by a wolf. The lamb flatters the wolf by asking him to play music louder and praising his music. The wolf becomes pleased by the flattery and plays loudly enough to attract the shepherd and the dogs, who catch the wolf and save the lamb. The shared gold annotation formulated the implicit major claim as: "flattery and pride can get you in trouble".

The quantitative profile of this story shows the mixed nature of the model performance. The average supporting-span F1 was relatively high at 0.675, but the average SBERT score was only 0.270 and the average BERTScore F1 was 0.284. This means that the models often located relevant story evidence but did not formulate the same explicit formulation as the gold annotation.

The clearest example is the implicit major claim. Gemma formulated the major claim as "Intelligence and quick thinking can overcome physical strength". Mistral formulated it as "Using clever tactics can help overcome a stronger adversary", and Qwen formulated it as "Quick thinking and cleverness can overcome physical strength and save one from danger". These formulations are plausible interpretations of the story from the lamb's perspective. The lamb does survive by using quick thinking and clever manipulation. However, the gold annotation focused on the wolf's weakness: flattery and pride lead to trouble. The models therefore identified a relevant narrative outcome but oriented the moral toward the cleverness of the victim rather than the vanity of the aggressor.

This difference is important because it shows that high supporting-span overlap does not guarantee correct implicit interpretation. A model can select the same

final story episode as the gold annotation, especially the rescue of the lamb and the capture of the wolf, but still derive a different moral from it. In this case, the span evidence supports both possible interpretations: the lamb's cleverness and the wolf's pride. The gold annotation selected the latter, while the models often selected the former.

The premises show the same pattern. The gold annotation included the premise "you get some compliments that might not be genuine", supported by the lamb telling the wolf that he plays better than the shepherd. It also included the premise "because of those disingenuous compliments, you feel flattered", supported by the wolf becoming pleased and playing louder. Mistral and Qwen produced formulations such as "Flattery can distract and overconfidence can lead to downfall" and "Pride and vanity can make powerful individuals vulnerable to manipulation". These are semantically closer to the gold interpretation, but they also compress the premise and the major claim into a broader moral statement. The models therefore captured the relevance of flattery but did not always preserve the same component structure as the gold annotation.

STORAL_008 is therefore useful because it shows a specific type of error: correct evidence but different implicit argument. This is especially relevant for the thesis because it demonstrates that implicit argument extraction is not only a span-identification task. The model must also decide which moral perspective the story supports. These stories are open to interpretation due to their narrative nature and subjectivity, therefore it is not always the case that the human annotators are right and the models are wrong. It all stems from the fact that the story can support multiple plausible morals.

STORAL_007: Mixed case with metric disagreement

STORAL_007 was selected as a mixed case because it showed relatively high SBERT and supporting-span alignment but lower BERTScore F1. The story describes a house-dog who is offered pieces of meat by a thief. The dog refuses to be fooled, remains loyal to his master, attacks the thief, and is later rewarded. The shared gold annotations included two implicit major claims: "true loyalty is stronger than temptation" and "staying loyal through temptations is rewarded". The gold premise was: "you are being bribed with something you love but you do not let that falter your loyalty easily".

This story had an average SBERT score of 0.607 and an average supporting-span F1 of 0.655, but an average BERTScore F1 of only 0.292. This pattern suggests that the model formulations were often semantically related to the gold annotations, even though the wording differed enough to reduce BERTScore. For example, Gemma formulated the first major claim as "Faithfulness and integrity are eventually rewarded". This is close to the gold claim "staying loyal through temptations is rewarded", although it uses different wording and introduces the broader concept of integrity. Similarly, Qwen formulated the second major claim as "True loyalty and dedication cannot be bought or compromised by bribes". This is strongly related to the gold annotation main claim, although it is more explicit about bribery.

The lower BERTScore values are therefore not always evidence of a poor interpretation. In several cases, the model preserved the underlying moral but expressed it with different lexical choices. This explains why STORAL_007 is useful as a metric-comparison case. SBERT appears more tolerant of paraphrase-level similarity, while BERTScore was lower when the model used different wording or condensed the gold interpretation into a more general moral statement.

The main remaining errors involved missed components and component-level compression. For example, Mistral missed the gold claim "staying loyal through temptations is rewarded", while Qwen produced an extra C-IMP annotation: "Steadfast

loyalty and protective behavior deserve recognition and reward". This extra annotation is not unreasonable, because the dog is rewarded at the end of the story. However, it does not correspond directly to the gold label structure. The model therefore produced a plausible additional interpretation, but not one that was part of the gold annotation.

The premise was also a difficult case. The gold premise focused on the dog being tempted with something he likes but refusing to let his loyalty become weaker. Gemma formulated this as "Loyalty involves prioritizing the safety of others over personal gain". Mistral formulated it as "A faithful dog will not be tempted by bribes", and Qwen formulated it as "Temptations are often used to undermine someone's commitment or duty". These are all related to the gold premise, but they vary in how specific they are. Mistral is closest to the exact story situation, while Gemma and Qwen generalize the premise into a broader statement about loyalty and temptation. This suggests that the models generally understood the overall idea, but they did not always express it at the same level of detail or in the same way as the gold formulation.

STORAL_005: Low-performing case

Lastly, STORAL_005 was selected as the low-performing case. The story describes a rabbit who leaves his home, a weasel who enters and refuses to leave, and a cat who offers to settle the dispute. The rabbit and the weasel trust the cat and come close to her, after which the cat catches them both. The shared gold annotation formulated the implicit major claim as: "trusting someone blindly can hurt everyone involved".

This story had low scores across all three evaluation dimensions. The average SBERT score was 0.293, the average BERTScore F1 was 0.243, and the average supporting-span F1 was 0.333. The main difficulty was that the models often inferred a different lesson from the story than the gold annotation. Instead of focusing on blind trust and mutual harm, the outputs focused on deception, conflict resolution, or property/security.

For example, Gemma formulated the major claim as "Conflict between two parties can be exploited by a third party for their own benefit". This is related to the story, because the cat does exploit the rabbit and the weasel, but it changes the moral focus. The gold annotation emphasizes the danger of trusting someone blindly, while Gemma emphasizes third-party exploitation. Mistral produced an even more different formulation: "Deception can be used to resolve conflicts when other methods fail". This formulation is problematic because it makes deception sound like an effective solution rather than a harmful act. Qwen however, produced the closest formulation: "Trusting a dangerous or untrustworthy third party to mediate a conflict can lead to mutual destruction". This is more aligned with the gold annotation, because it preserves the idea of misplaced trust and harm to both parties, although it is more specific than the gold formulation.

The implicit premises were also difficult for the models. The gold annotation included the premise "someone can intervene by trying to solve the disagreement", supported by the cat offering to settle the dispute. It also included the premise "when experiencing a disagreement and looking for a solution", supported by the conflict between the rabbit and the weasel. These premises were often missed or replaced by unrelated statements. For instance, Gemma produced "Negligence can lead to the loss of one's security", and Qwen produced "Neglecting to secure one's property creates opportunities for opportunistic intruders". These statements are based on the beginning of the story, where the rabbit forgets to latch the door, but they do not correspond to the gold argumentative structure. The models therefore focused on a different supporting reasoning: the rabbit's carelessness rather than the later problem of trusting the cat.

STORAL_005 illustrates one of the central challenges of implicit argument extraction from moral stories. A story can support more than one possible lesson. The models were able to produce story-relevant morals, but they did not consistently infer the same implicit argument that was selected in the gold annotation. This shows why semantic similarity metrics must be interpreted carefully: a low score does not necessarily mean that the model output is unrelated to the story, but it does mean that the model reconstructed a different argumentative interpretation.

Table 5.15: Representative examples from the STORAL error analysis.

Text	Label	Pattern	Example and interpretation
STORAL_011	MC-IMP	Meaningful near match	The gold formulation was "you can overcome your problems by facing them with courage". Model outputs such as "Facing your fears with courage is the most effective way to overcome them" preserved the same implicit meaning with different wording.
STORAL_011	P-IMP	Missed implicit premise	The premise "you are trying to escape your problems" was missed by all three models. The models recovered the final moral more successfully than the intermediate reasoning step.
STORAL_008	MC-IMP	Correct span, weak formulation	The models often selected the rescue/capture storyline, but formulated the moral as cleverness overcoming strength rather than flattery and pride causing trouble.
STORAL_008	P-IMP	Component compression	Some model outputs captured flattery and downfall in one broad statement, but did not preserve the distinction between the premise about compliments and the major claim about pride leading to trouble.
STORAL_007	MC-IMP	Metric disagreement	Outputs such as "True loyalty and dedication cannot be bought or compromised by bribes" were semantically close to the gold formulation about loyalty being stronger than temptation, even when BERTScore F1 was lower.
STORAL_007	C-IMP	Extra model annotation	Qwen produced an additional claim about loyalty deserving recognition and reward. This was plausible from the story ending, but it did not correspond to a gold C-IMP annotation.
STORAL_005	MC-IMP	Semantically weak formulation	The gold formulation focused on blind trust causing harm. Some model outputs instead focused on third-party exploitation or deception as conflict resolution, which shifted the moral of the story.
STORAL_005	P-IMP	Wrong causal focus	Some model outputs focused on the rabbit forgetting to latch the door, producing formulations about negligence or property security. These were story-related but did not correspond to the gold argument about trusting the cat.

Cross-text error patterns

Across the four STORAL stories, the most frequent error type was the missed implicit component. This occurred especially for implicit premises. The models often recovered the global moral of a story more successfully than the supporting reasoning steps. This suggests that LLMs are relatively good at producing a general lesson from a moral story, but less consistent at reconstructing the internal argumentative structure that links narrative events to that lesson.

A second major pattern was semantic weak formulation. In these cases, the model generated a plausible moral, but not the same moral as the gold annotation. STORAL_005 and STORAL_008 illustrate this especially clearly. In STORAL_005, the gold annotation focused on blind trust, while the models often focused on deception, conflict resolution, or carelessness. In STORAL_008, the gold annotation focused on flattery and pride, while the models often focused on cleverness overcoming strength. These outputs are not random, but they represent different interpretations of the same story.

A third pattern was evidence-meaning mismatch. STORAL_008 showed that the models could select a relevant supporting span but still produce a different explicit formulation. This indicates that supporting-span overlap alone is not enough for evaluating implicit argument extraction. The model may identify the right narrative moment while still drawing the wrong inference from it.

A fourth pattern was metric disagreement. STORAL_007 showed that SBERT and BERTScore do not always evaluate similarity in the same way. Some model formulations were semantically close to the gold annotation but used different wording, which led to relatively high SBERT similarity but lower BERTScore F1. This supports the decision to use both metrics in the STORAL evaluation, since each highlights a different aspect of similarity.

Finally, the STORAL error analysis shows that implicit argument extraction is affected by the interpretive openness of narrative texts. Moral stories often allow several possible lessons. The models frequently produced plausible interpretations, but these were not always the same interpretations selected in the gold annotation. This supports the argument of the thesis: narrativity affects LLM performance not only because arguments are implicit, but also because the same story events can be interpreted in different ways.

6

CONCLUSION

This thesis intended to answer the research question: How narrativity and discourse type influence the identification of explicit and implicit argumentative components by human annotators and Large Language Models. The study compared STORAL stories, where argumentative meaning is often implicit and embedded in narrative events, with TED talks, where argumentation is generally more explicit but still frequently supported by narrative elements.

The results show that narrativity affects argument identification by increasing interpretive complexity. In the TED corpus, the argumentation layer reached very high human-human agreement, with an average soft Gamma score of 0.900. This indicates that human annotators could identify explicit argumentative components, such as major claims, claims, and premises, in a consistent way. However, the narrative layer reached much lower human-human agreement, with an average soft Gamma score of 0.105. This suggests that identifying the rhetorical function of narrative spans was more ambiguous, since the same passage could serve several roles, such as background, evidence, introduction, or claim presentation.

The LLMs did not reach the level of human agreement in the TED argumentation layer. Although Gemma performed best for argumentation, with an average F1 of 0.330, the models often missed human annotations, selected overly broad spans, or confused major claims, claims, and premises. For the narrative layer, the models achieved somewhat higher F1 scores, with Qwen reaching 0.445 and Gemma 0.438, but these results should be interpreted carefully because the human reference itself was less stable. The qualitative analysis showed that the models often identified broad narrative sections, but struggled to segment them according to precise rhetorical function.

The STORAL results showed a different but related challenge. Because the argumentative meaning in these stories was usually implicit, the models had to infer and formulate the underlying argument. They were often able to produce plausible moral interpretations, especially for implicit major claims, but they were less successful at reconstructing the supporting premises that connected story events to those claims. This shows that LLMs can often identify the general lesson of a narrative, but have more difficulty recovering its internal argumentative structure.

Overall, the research question can be answered as follows: narrativity and discourse type influence argument identification by affecting both the explicitness of argumentative components and the amount of interpretation required. TED talks allowed more stable human annotation of explicit argument components, but still challenged LLMs at the level of span selection and label assignment. STORAL stories required stronger inference because their argumentative meaning was narratively embedded, which made model outputs more open to variation. The thesis therefore shows that narrative matters for argument extraction: it does not only change how arguments are expressed, but also how humans and models identify and interpret them.

6.1 LIMITATIONS AND FUTURE WORK

Although this study provides useful insights, its findings should be interpreted in light of several methodological limitations. The first concerns the size and scope of the dataset. The study worked with a relatively small corpus of TED talks and STORAL stories, which made detailed manual annotation and qualitative error analysis possible. This was important because the task required close interpretation of argumentative and narrative structures. However, the small size also means that the findings should not be treated as fully generalizable across all public talks, moral stories, or narrative argumentative texts. Future work could therefore apply the same annotation framework to a larger number of texts, or to additional genres such as opinion articles, political speeches, personal essays, debates, or social media discussions. This would make it possible to test whether the patterns observed here are specific to TED and STORAL, or whether they reflect broader tendencies in how narrative shapes argumentative meaning.

A second limitation regards the interpretive nature of the annotation task itself. The argumentation layer, especially in the TED corpus, reached strong human-human agreement, which suggests that explicit argumentative components could be identified with reasonable consistency. The narrative layer, however, produced much lower agreement between human annotators. This does not mean that the layer was unhelpful, but it does show that the rhetorical-function labels *Exordium*, *Narratio*, *Propositio*, *Confirmatio*, and *Peroratio* were difficult to apply consistently. One reason for this is that narrative passages in TED talks often perform several functions at the same time. A personal anecdote, for example, can introduce the topic, provide background, and function as evidence for a claim. Future work could refine this part of the annotation scheme by providing more detailed decision rules, adding more examples, or allowing multi-label annotation when a narrative span has more than one rhetorical function.

The STORAL annotation process also involved an important methodological choice. Because implicit argumentative meaning is highly interpretive, the final STORAL reference annotations were created as a shared gold standard through discussion between the annotators. This made the evaluation more consistent, but it also reduced the visibility of alternative interpretations. The error analysis showed that some LLM outputs did not match the gold formulation of the human annotation, but were still possible conclusions of the same story. For example, a model might infer a lesson about cleverness, while the gold annotation focused on pride or blind trust. This suggests that future work could use an annotation approach that is more focused on differing perspectives, where multiple valid interpretations are considered valid instead of being resolved into a single gold annotation. This would be especially useful for moral narratives, where the same story can support more than one possible argumentative interpretation.

The evaluation methods also place limits on how the results can be interpreted. For TED, Gamma and F1-based measures were useful for comparing span and label similarity, but they cannot fully capture cases where a model selected a broader or narrower span that was still semantically related to the human annotation. For STORAL, SBERT and BERTScore made it possible to compare implicit formulations semantically, but the qualitative analysis showed that the two metrics sometimes emphasized different aspects of similarity. SBERT was more tolerant of broader paraphrases, while BERTScore was more sensitive to wording and lexical overlap. This means that semantic similarity metrics are useful, but they cannot replace manual interpretation when the task involves implicit argumentative meaning. Future evaluations could combine automatic scores with human judgments of whether a model output is not only semantically similar, but also argumentatively equivalent to the reference annotation.

Another limitation concerns the use of prompt-based LLM annotation. The models were not fine-tuned for the task, and their performance depended strongly on the design of the prompts. Different tasks required different prompting strategies: TED argumentation benefited from few-shot examples, TED narration performed best with detailed definitions but no examples, and STORAL required examples because the task involved both selecting supporting spans and generating explicit formulations. This shows that the results reflect model behavior under specific prompting conditions, rather than the maximum possible performance of these models. Future research could compare prompt-based annotation with fine-tuned or supervised models trained on human annotations. Such a comparison would help determine whether the errors observed in this thesis are mainly caused by the models' general limitations, by the prompts, or by the difficulty of the annotation task itself.

Finally, the study compared three open-weight instruction-tuned models of broadly similar size. This made it possible to compare behavior across different model families, but it does not represent the full range of available LLMs. Larger models, models specifically trained for discourse analysis, or models fine-tuned on argument-mining datasets might produce different results. Future work could therefore expand the model comparison and investigate whether the same error patterns appear across different model scales and architectures.

Taken together, these limitations point toward a broader direction for future research: argument extraction from narrative should not be treated only as a technical extraction problem. It is also an interpretive task involving discourse structure, implicit meaning, and sometimes multiple valid readings. Future work should therefore continue combining quantitative evaluation with qualitative error analysis. This is especially relevant to Digital Humanities research, where computational methods are useful not because they remove interpretation, but because they make it possible to examine interpretive patterns systematically.

BIBLIOGRAPHY

- Chen, B., Zhang, Z., Langrené, N., and Zhu, S. (2025). Unleashing the potential of prompt engineering for large language models. *Patterns*, 6(6).
- Chen, Y., Wen, Z., Fan, G., Chen, Z., Wu, W., Liu, D., Li, Z., Liu, B., and Xiao, Y. (2023). Mapo: Boosting large language model performance with model-adaptive prompt optimization. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 3279–3304.
- Corral Jr, M. (2020). Ted talks transcripts for nlp. <https://www.kaggle.com/datasets/miguelcorraljr/ted-ultimate-dataset>.
- Da San Martino, G., Barrón-Cedeño, A., Wachsmuth, H., Petrov, R., and Nakov, P. (2020). Semeval-2020 task 11: Detection of propaganda techniques in news articles. In *Proceedings of the fourteenth workshop on semantic evaluation*, pages 1377–1414.
- Falk, N. and Lapesa, G. (2023). Storyarg: a corpus of narratives and personal experiences in argumentative texts. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2350–2372.
- Guan, J., Liu, Z., and Huang, M. (2022). A corpus for understanding and generating moral stories. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 5069–5087.
- James, J. (2026). Counting on consensus: Selecting the right inter-annotator agreement metric for nlp annotation and evaluation. *arXiv preprint arXiv:2603.06865*.
- Klie, J.-C., Bugert, M., Boullosa, B., de Castilho, R. E., and Gurevych, I. (2018). The inception platform: Machine-assisted and knowledge-oriented interactive annotation. In *Proceedings of the 27th International Conference on Computational Linguistics: System Demonstrations*, pages 5–9. Association for Computational Linguistics. Event Title: The 27th International Conference on Computational Linguistics (COLING 2018).
- Li, Y. (2023). A practical survey on zero-shot prompt design for in-context learning. In *Proceedings of the 14th international conference on recent advances in natural language processing*, pages 641–647.
- Mathet, Y., Widlöcher, A., and Métivier, J.-P. (2015). The unified and holistic method gamma (γ) for inter-annotator agreement measure and alignment. *Computational Linguistics*, 41(3):437–479.
- Mazzocco, P. J. and Green, M. C. (2011). Narrative persuasion in legal settings: What’s the story. *The Jury Expert*, 23(3):27–34.
- Nabhani, S., Pianzola, F., Al-Khatib, K., and Nissim, M. (2026). Argus: Seeing the influence of narrative features on persuasion in argumentative texts. *arXiv preprint arXiv:2602.24109*.
- Pauli, A. B., Augenstein, I., and Assent, I. (2025). Measuring and benchmarking large language models’ capabilities to generate persuasive language. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 10056–10075.

- Pianzola, F. (2018). Looking at narrative as a complex system: The proteus principle. In *Narrating complexity*, pages 101–122. Springer.
- Powers, D. M. (2020). Evaluation: from precision, recall and f-measure to roc, informedness, markedness and correlation. *arXiv preprint arXiv:2010.16061*.
- Purdue, U. (2023). Classical argument. https://owl.purdue.edu/owl/general_writing/academic_writing/historical_perspectives_on_argumentation/classical_argument.html.
- Reimers, N. and Gurevych, I. (2019). Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pages 3982–3992.
- Riad, R., Titeux, H., and Favre, L. (2020). The soft-gamma-agreement: An alternate measure to gamma. <https://github.com/bootphon/pygamma-agreement/blob/master/docs/source/soft-gamma.rst>. Documentation file from the pygamma-agreement GitHub repository.
- Stab, C. and Gurevych, I. (2014a). Annotating argument components and relations in persuasive essays. In *Proceedings of COLING 2014, the 25th international conference on computational linguistics: Technical papers*, pages 1501–1510.
- Stab, C. and Gurevych, I. (2014b). Identifying argumentative discourse structures in persuasive essays. In *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*, pages 46–56.
- Stab, C. and Gurevych, I. (2017). Parsing argumentation structures in persuasive essays. *Computational Linguistics*, 43(3):619–659.
- Tian, Y., Huang, T., Liu, M., Jiang, D., Spangher, A., Chen, M., May, J., and Peng, N. (2024). Are large language models capable of generating human-level narratives? In Al-Onaizan, Y., Bansal, M., and Chen, Y.-N., editors, *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 17659–17681, Miami, Florida, USA. Association for Computational Linguistics.
- Tindale, C. (2017). Narratives and the concept of argument. In *Narration as argument*, pages 11–30. Springer.
- Titeux, H. and Riad, R. (2021). pygamma-agreement: Gamma γ measure for inter/intra-annotator agreement in python. *Journal of Open Source Software*, 6(62):2989.
- Zhang, T., Kishore, V., Wu, F., Weinberger, K. Q., and Artzi, Y. (2019). Bertscore: Evaluating text generation with bert. *arXiv preprint arXiv:1904.09675*.
- Üzelgün, M. A., Aktar, M., Turner, H., and Oruç, R. (2017). Narrative and argumentative: Towards a joint framework for examining everyday discourse.

A

PROMPT TEMPLATES

This appendix contains the prompt templates used for the LLM-based annotation experiments. The prompts are grouped by dataset, annotation layer, and development stage.

A.1 FINAL TED NARRATIVE-LAYER PROMPT (ZERO-SHOT)

```
1 system_prompt_narrative_final = """
2 You are an expert annotator trained in narrative discourse analysis and argumentation
   mining.
3
4 Your task is to identify narrative spans in TED Talk transcripts and classify their
   rhetorical function within the broader argument.
5
6 Return ONLY valid JSON.
7 Do not explain your reasoning.
8 Do not use markdown.
9 """
10
11 def build_narrative_prompt_final(ted_text):
12     return f"""
13 TASK:
14 Annotate narrative spans in the TED Talk.
15
16 A narrative span is a segment of text that includes story-like elements such as:
17 - events,
18 - actions,
19 - situations,
20 - people or characters,
21 - temporal progression,
22 - causal progression,
23 - descriptions of what happened,
24 - examples or anecdotes.
25
26 Only annotate narrative spans that contribute to the discourse.
27
28 Use ONLY these labels based on their definitions:
29
30 1. Exordium
31 A narrative opening that engages the audience, introduces the topic, or draws attention.
32 Usually appears near the beginning, but label by function, not position.
33
34 2. Narratio
35 Narrative background or context that helps the audience understand the issue.
36 Use this for situations, historical background, prior events, or conditions that explain
   why the topic matters.
37
38 3. Propositio
39 A narrative span that introduces or frames the speaker's central stance, project,
   purpose, or direction of the argument.
40 Use this when the narrative helps present what the speaker is arguing for or what the
   talk will develop.
```

```

41
42 4. Confirmatio
43 A narrative span that functions as evidence, support, illustration, example, case study,
    or proof for the speaker's argument. In some cases, narrative evidence may also
    implicitly challenge alternative viewpoints by presenting contrasting outcomes
44 Use this for concrete events or examples that support a claim.
45
46 6. Peroratio
47 A narrative closing that summarizes, reinforces, or leaves a final impression supporting
    the overall argument.
48 Usually appears near the end, but label by function, not position.
49
50 7. Other
51 A narrative span that contributes to the discourse but does not clearly fit the labels
    above.
52 Use this sparingly.
53
54 IMPORTANT DECISION RULES:
55
56 - Label by rhetorical function, not by location.
57 - A span is not NARRATIO only because it appears early.
58 - A span is not PERORATIO only because it appears at the end.
59 - A personal anecdote is not automatically argumentative.
60 - A description is not automatically narrative evidence.
61 - Only annotate spans that have a clear narrative function in the talk.
62 - If a passage is purely argumentative with no narrative elements, do not annotate it.
63 - If a passage is only a greeting, joke, transition, or filler, do not annotate it.
64 - If unsure, do not annotate.
65
66 SPAN RULES:
67
68 - Copy exact text spans from the TED Talk.
69 - Do not paraphrase.
70 - Do not invent spans.
71 - Select complete narrative units, not isolated words.
72 - Prefer one coherent narrative segment over many tiny fragments.
73 - Do not include unnecessary surrounding text.
74 - Do not duplicate the same span with multiple labels.
75 - Choose the single best functional label.
76
77 IMPORTANT
78 - Most spans have more than one sentence, to which only one single label is assigned.
79 - A text doesn't necessarily have to include all of the narrative labels.
80
81
82 TED TALK:
83 \{"\"
84 {ted_text}
85 \{"\"
86 ""

```

A.2 LAST TESTED TED NARRATIVE-LAYER PROMPT (ZERO-SHOT)

```

1 system_prompt_narrative_last = ""
2 You are an expert annotator trained in narrative discourse analysis and argumentation
    mining.
3

```



```

4 Your task is to identify narrative spans in TED Talk transcripts and classify their
   rhetorical function within the broader argument.
5
6 Return ONLY valid JSON.
7 Do not explain your reasoning.
8 Do not use markdown.
9 """
10
11 def build_narrative_prompt_last(ted_text):
12     return f"""
13 TASK:
14 Annotate narrative spans in the TED Talk.
15
16 A narrative span is a segment of text that includes story-like elements such as:
17 - events,
18 - actions,
19 - situations,
20 - people or characters,
21 - temporal progression,
22 - causal progression,
23 - descriptions of what happened,
24 - examples or anecdotes.
25
26 Only annotate narrative spans that contribute to the discourse.
27
28 Use ONLY these labels based on their definitions:
29
30 1. Exordium
31 A narrative opening that engages the audience, introduces the topic, or draws attention.
32 Usually appears near the beginning, but label by function, not position.
33
34 2. Narratio
35 Narrative background or context that helps the audience understand the issue.
36 Use this for situations, historical background, prior events, or conditions that explain
   why the topic matters.
37
38 3. Propositio
39 A narrative span that introduces or frames the speaker's central stance, project,
   purpose, or direction of the argument.
40 Use this when the narrative helps present what the speaker is arguing for or what the
   talk will develop.
41
42 4. Confirmatio
43 A narrative span that functions as evidence, support, illustration, example, case study,
   or proof for the speaker's argument. In some cases, narrative evidence may also
   implicitly challenge alternative viewpoints by presenting contrasting outcomes
44 Use this for concrete events or examples that support a claim.
45
46 6. Peroratio
47 A narrative closing that summarizes, reinforces, or leaves a final impression supporting
   the overall argument.
48 Usually appears near the end, but label by function, not position.
49
50 7. Other
51 A narrative span that contributes to the discourse but does not clearly fit the labels
   above.
52 Use this sparingly.
53
54 IMPORTANT DECISION RULES:
55
56 - Label by rhetorical function, not by location.
57 - A span is not NARRATIO only because it appears early.
58 - A span is not PERORATIO only because it appears at the end.
59 - A personal anecdote is not automatically argumentative.
60 - A description is not automatically narrative evidence.
61 - Only annotate spans that have a clear narrative function in the talk.
62 - If a passage is purely argumentative with no narrative elements, do not annotate it.
63 - If a passage is only a greeting, joke, transition, or filler, do not annotate it.
64 - If unsure, do not annotate.
65

```

```

66 SPAN RULES:
67
68 - Copy exact text spans from the TED Talk.
69 - Do not paraphrase.
70 - Do not invent spans.
71 - Select complete narrative units, not isolated words.
72 - Prefer one coherent narrative segment over many tiny fragments.
73 - Do not include unnecessary surrounding text.
74 - Do not duplicate the same span with multiple labels.
75 - Choose the single best functional label.
76
77 IMPORTANT
78 - Most spans have more than one sentence, to which only one single label is assigned.
79 - A text doesn't necessarily have to include all of the narrative labels.
80
81
82 TED TALK:
83 \{"\"
84 {ted_text}
85 \{"\"
86 ""

```

A.3 FINAL TED ARGUMENTATION-LAYER PROMPT (FEW-SHOT)

```

1 system_prompt_argumentation_final = ""
2 You are an expert annotator trained in argumentation mining.
3 Your task is to annotate argumentative components in TED Talk transcripts.
4 Return ONLY valid JSON.
5 Do not explain your reasoning.
6 ""
7
8 def build_argumentation_prompt_final(ted_text):
9     return f"""
10 Identify all argumentation elements in the following TED Talk transcripts.
11 An argument is made of 3 argument components listed below.
12
13 Use ONLY these labels based on their definitions:
14
15 - MC-EXP: The major claim represents the central message or overarching conclusion of a
    text. The major claim represents the stance of the author about the essay topic. It
    is also called a thesis statement and frequently indicated by opinion expressions
    like "From my point of view...", "In my opinion...", "I strongly believe that...",
    etc. Usually, the major claim is present in the introduction or conclusion of an
    essay or in both.
16
17 - C-EXP: Claims represent intermediate statements that contribute to the development of
    the major claim. They are typically evaluative or interpretive in nature and
    express a viewpoint that requires justification.
18
19 - P-EXP: Premises provide the reasoning or evidence that supports a claim. They play a
    crucial role in establishing the plausibility of an argument. Premises can also
    proceed after indicators such as "because...", "for example...", "moreover...", etc.
20
21
22 IMPORTANT:
23 Most sentences in TED Talks are NOT argumentative.

```

```

24
25 Do NOT annotate:
26
27 - storytelling,
28 - greetings,
29 - jokes,
30 - rhetorical questions,
31 - transitions,
32 - descriptions,
33 - facts,
34 - background information,
35 - emotional language,
36 - personal anecdotes unless they directly support a claim.
37
38 Annotate ONLY spans that clearly function as argumentation.
39
40 RULES:
41
42 Copy EXACT text spans.
43 Select the MINIMAL meaningful span.
44 Do NOT paraphrase.
45 Do NOT invent spans.
46 Do NOT annotate weak or uncertain cases.
47 If unsure, do NOT annotate.
48 Precision is more important than recall.
49
50 Example 1:
51 Text:
52 "I teach chemistry."
53
54 Output:
55 {{
56 "annotations": []
57 }}
58
59 Example 2:
60 Text:
61 "And they suppress institutions and destroy them so that they don't have to be held
    accountable. So do you have a heavily militarized country? And this leads to point
    four, what I call human cruelty."
62
63 Output:
64 {{
65 "annotations": [
66   {{
67     "label": "C-EXP",
68     "text": "they suppress institutions and destroy them so that they don't have to be
        held accountable"
69   }},
70   {{
71     "label": "P-EXP",
72     "text": "this leads to point four, what I call human cruelty"
73   }}
74 ]
75 }}
76
77 Example 3:
78 Text:
79 "Well, I believe that if we want to change what our cities look like, then we really
    have to change the decision-making processes that have given us the results that we
    have right now. We need a participation revolution, and we need it fast. The idea
    of voting as our only exercise in citizenship does not make sense anymore. People
    are tired of only being treated as empowered individuals every few years when it's
    time to delegate that power to someone else."
80
81 Output:
82 {{
83 "annotations": [
84   {{
85     "label": "MC-EXP",

```

```

86     "text": "if we want to change what our cities look like, then we really have to
87     change the decision-making processes that have given us the results that we have
88     right now"
89     },
90     {
91       "label": "C-EXP",
92       "text": "We need a participation revolution"
93     },
94     {
95       "label": "C-EXP",
96       "text": "we need it fast"
97     },
98     {
99       "label": "C-EXP",
100      "text": "The idea of voting as our only exercise in citizenship does not make
101      sense anymore"
102    },
103    {
104      "label": "P-EXP",
105      "text": "People are tired of only being treated as empowered individuals every few
106      years when it's time to delegate that power to someone else"
107    }
108  ]
109 }
110
111 Example 4:
112 "Thank you very much."
113
114 Output:
115 {
116   "annotations": []
117 }
118
119 Example 5:
120 Text:
121 "I blog and thousands of people read it."
122
123 Output:
124 {
125   "annotations": []
126 }
127
128 OUTPUT FORMAT:
129 {
130   "annotations": [
131     {
132       "label": "C-EXP",
133       "text": "exact copied span"
134     }
135   ]
136 }
137
138 TED Talk:
139 \"\"\"
140 {ted_text}
141 \"\"\"

```

A.4 LAST TESTED TED ARGUMENTATION-LAYER PROMPT (FEW-SHOT)

```

1 system_prompt_argumentation_last = """
2 You are an expert annotator trained in argumentation mining.
3 Your task is to annotate argumentative components in TED Talk transcripts.
4 Return ONLY valid JSON.
5 Do not explain your reasoning.
6 """
7
8 def build_argumentation_prompt_last(ted_text):
9     return f"""
10 Identify all argumentation elements in the following TED Talk transcripts.
11 An argument is made of 3 argument components listed below.
12
13 Use ONLY these labels based on their definitions:
14
15 - MC-EXP: The major claim represents the central message or overarching conclusion of a
      text. The major claim represents the stance of the author about the essay topic. It
      is also called a thesis statement and frequently indicated by opinion expressions
      like "From my point of view...", "In my opinion...", "I strongly believe that...",
      etc. Usually, the major claim is present in the introduction or conclusion of an
      essay or in both.
16
17 - C-EXP: Claims represent intermediate statements that contribute to the development of
      the major claim. They are typically evaluative or interpretive in nature and
      express a viewpoint that requires justification.
18
19 - P-EXP: Premises provide the reasoning or evidence that supports a claim. They play a
      crucial role in establishing the plausibility of an argument.
20
21
22 IMPORTANT:
23 Most sentences in TED Talks are NOT argumentative.
24
25 Do NOT annotate:
26
27 - storytelling,
28 - greetings,
29 - jokes,
30 - rhetorical questions,
31 - transitions,
32 - descriptions,
33 - facts,
34 - background information,
35 - emotional language,
36 - personal anecdotes unless they directly support a claim.
37
38 Annotate ONLY spans that clearly function as argumentation.
39
40 RULES:
41
42 Copy EXACT text spans.
43 Select the MINIMAL meaningful span.
44 Do NOT paraphrase.
45 Do NOT invent spans.
46 Do NOT annotate weak or uncertain cases.
47 If unsure, do NOT annotate.
48 Precision is more important than recall.
49
50 Example 1:
51 Text:
52 "When I was six years old, my father took me fishing."
53
54 Output:
55 {{
56 "annotations": []

```

```

57 }}
58
59 Example 2:
60 Text:
61 "These sympathetic smiles spread all throughout the room now, and I realize that all of
    my other students are in the same boat, that maybe they took notes or memorized
    definitions from the textbook, but not one of them really understood the main ideas
    ."
62
63 Output:
64 {{
65   "annotations": [
66     {{
67       "label": "C-EXP",
68       "text": "not one of them really understood the main ideas"
69     }},
70     {{
71       "label": "P-EXP",
72       "text": "all of my other students are in the same boat, that maybe they took notes
    or memorized definitions from the textbook"
73     }}
74   ]
75 }}
76
77 Example 3:
78 Text:
79 "The range of digital relations is extraordinary. In the context of this broad range of
    digital relations, safely seeking strangeness might very well be a new basis for
    that innovation."
80
81 Output:
82 {{
83   "annotations": [
84     {{
85       "label": "MC-EXP",
86       "text": "In the context of this broad range of digital relations, safely seeking
    strangeness might very well be a new basis for that innovation"
87     }},
88     {{
89       "label": "C-EXP",
90       "text": "The range of digital relations is extraordinary"
91     }}
92   ]
93 }}
94
95 Example 4:
96 "Thank you all for being here today."
97
98 Output:
99 {{
100  "annotations": []
101 }}
102
103 OUTPUT FORMAT:
104 {{
105   "annotations": [
106     {{
107       "label": "C-EXP",
108       "text": "exact copied span"
109     }}
110   ]
111 }}
112
113
114 Output format:
115 {{
116   "annotations": [
117     {{
118       "label": "C-EXP",
119

```

```

120     "text": "exact copied span"
121     }}
122 ]
123 }}
124
125 TED Talk:
126 \"\\"
127 {ted_text}
128 \"\\"
129 ""

```

A.5 FINAL STORAL IMPLICIT-ARGUMENTATION PROMPT (FEW-SHOT)

```

1  system_prompt_implicit_final = ""
2  You are an expert annotator trained in implicit argumentation mining.
3
4  Your task is to identify implicit argumentative components in texts.
5
6  Implicit components are NOT directly stated in the text.
7  They must be inferred from context.
8
9  Return ONLY valid JSON.
10 Do not explain your reasoning.
11 Do not use markdown.
12 ""
13 def build_implicit_prompt_final(storal_text):
14
15     return f""
16
17 TASK:
18 Identify implicit argumentative components in the story text.
19
20 An implicit component is: NOT explicitly stated, but strongly implied by the text,
    necessary to understand the argumentative meaning, and supported by the text as a
    whole.
21
22 For every implicit component you MUST provide:
23 - label
24 - supporting span
25 - explicit formulation
26
27 IMPORTANT:
28 The supporting span is NOT the implicit component itself.
29 The supporting span is the text segment from which the implicit meaning is inferred.
30
31 Use ONLY these labels based on their definitions:
32
33 MC-IMP: Implicit major claim is the overall inferred takeaway, moral, lesson, central
    message, or broad conclusion of the text.
34 C-IMP: Implicit claim: is an inferred intermediate viewpoint, interpretation, evaluative
    meaning, or unstated argumentative step.
35 P-IMP: Implicit premise is an unstated reasoning step, assumption, or inferred
    justification needed to connect ideas.
36
37 STRICT ANNOTATION RULES:
38
39 Only annotate implications that are STRONGLY supported.

```

```

40 Do NOT annotate speculative interpretations.
41 Do NOT invent morals unsupported by the text.
42 Do NOT generate multiple possible interpretations.
43 Select ONLY the best-supported interpretation.
44 The explicit formulation must stay close to the text.
45 The explicit formulation must be generalizable.
46 The explicit formulation must be a COMPLETE sentence.
47 The explicit formulation must NOT simply restate the supporting span.
48 If the inferred meaning is weak or ambiguous, do NOT annotate it.
49 If unsure, do NOT annotate.
50 Precision is MUCH more important than recall.
51
52 SUPPORTING SPAN RULES:
53
54 Copy EXACT text from the TED Talk.
55 Do NOT paraphrase the supporting span.
56 Select the MINIMAL span necessary for the inference.
57 Do NOT select the entire TED Talk.
58 Do NOT include unrelated surrounding text.
59 The supporting span should directly motivate the inferred meaning.
60
61 EXPLICIT FORMULATION RULES:
62
63 GOOD:
64
65 "Selfish behavior can harm others."
66 "Urban neglect damages public life."
67 "Experiencing consequences can teach responsibility."
68
69 BAD:
70
71 vague statements,
72 unsupported morals,
73 overly philosophical interpretations,
74 interpretations adding new ideas not grounded in the text.
75
76 NEGATIVE EXAMPLES:
77
78 Example 1
79
80 Text:
81 "It was raining heavily that morning."
82
83 Output:
84 {{
85 "annotations": []
86 }}
87
88 Example 2
89
90 Text:
91 "Thank you very much for inviting me here today."
92
93 Output:
94 {{
95 "annotations": []
96 }}
97
98 Example 3
99
100 Text:
101 "The bridge was built in 1952."
102
103 Output:
104 {{
105 "annotations": []
106 }}
107
108 POSITIVE EXAMPLES:
109

```



```

110 Example 4
111
112 Text:
113 "The hen and the grasshopper followed the advice of the dog and at the end of the year
    they had the necessary money for the cow. They also make a better living through
    their saving."
114
115 Output:
116 {{
117   "annotations": [
118     {{
119       "label": "P-IMP",
120       "supporting_span": "The hen and the grasshopper followed the advice of the dog",
121       "explicit_formulation": "You follow advice from a wise individual."
122     }},
123     {{
124       "label": "MC-IMP",
125       "supporting_span": "at the end of the year they had the necessary money for the
    cow. They also make a better living through their saving",
126       "explicit_formulation": "Hard work, responsible behavior and perseverance
    eventually leads to a better life."
127     }}
128   ]
129 }}
130
131 Example 5
132
133 Text:
134 "rather, you can just have a piece of leather cut in appropriate shape to cover your
    feet? The king was very much surprised by his suggestion and applauded the minister.
    He ordered for a pair of leather shoes for him and requested the countrymen to
    wear shoes."
135
136 Output:
137 {{
138   "annotations": [
139     {{
140       "label": "MC-IMP",
141       "supporting_span": "rather, you can just have a piece of leather cut in
    appropriate shape to cover your feet",
142       "explicit_formulation": "It is easier to adapt than to try to change the whole
    world"
143     }},
144     {{
145       "label": "P-IMP",
146       "supporting_span": "The king was very much surprised by his suggestion and
    applauded the minister. He ordered for a pair of leather shoes for him and
    requested the countrymen to wear shoes",
147       "explicit_formulation": "You find the simplest of ideas that turns out to be the
    most effective and least harmful"
148     }}
149   ]
150 }}
151
152 Example 6
153
154 Text:
155 ""Now you understand?? are you sorry that because of your selfishness I got beaten by
    farmer.."Fox cried, "Yes.. yes.. I have realized my mistake.. now please forgive me
    ..""
156
157 Output:
158 {{
159   "annotations": [
160     {{
161       "label": "MC-IMP",
162       "supporting_span": ""Now you understand?? are you sorry that because of your
    selfishness I got beaten by farmer..",
163       "explicit_formulation": "selfish actions can eventually bring consequences upon
    oneself"
    }},
  ]
}}

```

```

164     {{
165         "label": "P-IMP",
166         "supporting_span": "Yes.. yes.. I have realized my mistake.. now please forgive me
167         ..",
168         "explicit_formulation": "You get a taste of your own medicine and it teaches you an
169         important lesson"
170     }}
171 ]
172 }}
173 A text may contain:
174 - only one implicit major claim
175 OUTPUT FORMAT:
176 {{
177     "annotations": [
178         {{
179             "label": "MC-IMP",
180             "supporting_span": "exact copied text span",
181             "explicit_formulation": "clear inferred statement"
182         }}
183     ]
184 }}
185
186 Story:
187 \"\"\"
188 {storal_text}
189 \"\"\"
190
191 """

```

A.6 LAST TESTED STORAL IMPLICIT-ARGUMENTATION PROMPT (FEW-SHOT)

```

1  system_prompt_implicit_last = """
2  You are an expert annotator trained in implicit argumentation mining.
3
4  Your task is to identify implicit argumentative components in texts.
5
6  Implicit components are NOT directly stated in the text.
7  They must be inferred from context.
8
9  Return ONLY valid JSON.
10 Do not explain your reasoning.
11 Do not use markdown.
12 """
13 def build_implicit_prompt_last(storal_text):
14
15     return f"""
16
17 TASK:
18 Identify implicit argumentative components in the story text.
19
20 An implicit component is: NOT explicitly stated, but strongly implied by the text,
21     necessary to understand the argumentative meaning, and supported by the text as a
22     whole.
23
24 For every implicit component you MUST provide:

```

```

23 - label
24 - supporting span
25 - explicit formulation
26
27 IMPORTANT:
28 The supporting span is NOT the implicit component itself.
29 The supporting span is the text segment from which the implicit meaning is inferred.
30
31 Use ONLY these labels based on their definitions:
32
33 MC-IMP: Implicit major claim is the overall inferred takeaway, moral, lesson, central
    message, or broad conclusion of the text.
34 C-IMP: Implicit claim: is an inferred intermediate viewpoint, interpretation, evaluative
    meaning, or unstated argumentative step.
35 P-IMP: Implicit premise is an unstated reasoning step, assumption, or inferred
    justification needed to connect ideas.
36
37 STRICT ANNOTATION RULES:
38
39 Only annotate implications that are STRONGLY supported.
40 Do NOT annotate speculative interpretations.
41 Do NOT invent morals unsupported by the text.
42 Do NOT generate multiple possible interpretations.
43 Select ONLY the best-supported interpretation.
44 The explicit formulation must stay close to the text.
45 The explicit formulation must be generalizable.
46 The explicit formulation must be a COMPLETE sentence.
47 The explicit formulation must NOT simply restate the supporting span.
48 If the inferred meaning is weak or ambiguous, do NOT annotate it.
49 If unsure, do NOT annotate.
50 Precision is MUCH more important than recall.
51
52 SUPPORTING SPAN RULES:
53
54 Copy EXACT text from the TED Talk.
55 Do NOT paraphrase the supporting span.
56 Select the MINIMAL span necessary for the inference.
57 Do NOT select the entire TED Talk.
58 Do NOT include unrelated surrounding text.
59 The supporting span should directly motivate the inferred meaning.
60
61 EXPLICIT FORMULATION RULES:
62
63 GOOD:
64
65 "Selfish behavior can harm others."
66 "Urban neglect damages public life."
67 "Experiencing consequences can teach responsibility."
68
69 BAD:
70
71 vague statements,
72 unsupported morals,
73 overly philosophical interpretations,
74 interpretations adding new ideas not grounded in the text.
75
76 NEGATIVE EXAMPLES:
77
78 Example 1
79
80 Text:
81 "It was raining heavily that morning."
82
83 Output:
84 {{
85 "annotations": []
86 }}
87
88 Example 2
89

```

```

90 Text:
91 "Thank you very much for inviting me here today."
92
93 Output:
94 {{
95   "annotations": []
96 }}
97
98 Example 3
99
100 Text:
101 "The bridge was built in 1952."
102
103 Output:
104 {{
105   "annotations": []
106 }}
107
108 POSITIVE EXAMPLES:
109
110 Example 4
111
112 Text:
113 "The trees gave the man branches, and he later used them to build an axe and cut down
    the trees."
114
115 Output:
116 {{
117   "annotations": [
118     {{
119       "label": "MC-IMP",
120       "supporting_span": "The trees gave the man branches, and he later used them to
        build an axe and cut down the trees",
121       "explicit_formulation": "Helping an enemy can contribute to one's own destruction
        ."
122     }}
123   ]
124 }}
125
126 Example 5
127
128 Text:
129 "Fox ignored Camel's warning, caused trouble, and later suffered the consequences
    herself."
130
131 Output:
132 {{
133   "annotations": [
134     {{
135       "label": "P-IMP",
136       "supporting_span": "Fox ignored Camel's warning, caused trouble, and later
        suffered the consequences herself",
137       "explicit_formulation": "Experiencing consequences can help individuals recognize
        their mistakes."
138     }}
139   ]
140 }}
141
142 Example 6
143
144 Text:
145 "The river had gradually become polluted, covered, and transformed into a trash-filled
    space."
146
147 Output:
148 {{
149   "annotations": [
150     {{
151       "label": "C-IMP",

```

```

152     "supporting_span": "The river had gradually become polluted, covered, and
153     transformed into a trash-filled space",
154     "explicit_formulation": "Neglecting urban environments can damage community life."
155   }}
156 }
157
158 A text may contain:
159 - only one implicit major claim,
160 - or one implicit premise,
161
162 OUTPUT FORMAT:
163 {{
164   "annotations": [
165     {{
166       "label": "MC-IMP",
167       "supporting_span": "exact copied text span",
168       "explicit_formulation": "clear inferred statement"
169     }}
170   ]
171 }}
172
173
174 Story:
175 \\\\"
176 {storal_text}
177 \\\\"
178 ""

```